Vision Studio: Landmark-Masked Facial Inpainting with DSPy Prompt Rewriting

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"Vision Studio: Landmark-Masked Facial Inpainting with DSPy Prompt Rewriting"

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DECLARATION

It is to declare that this project is our original work. No part of this work has been submitted elsewhere, partially or entirely, for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor.

Relevant previous works presented in this report have been adequately acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

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Abstract

Diffusion models have made photorealistic facial editing widely available, but two problems limit how far such systems can be trusted. They drift a subject's identity while applying an edit, and the quality of an edit depends heavily on how the request happens to be worded. Vision Studio is a region-masked diffusion inpainting system for cosmetic face editing, in which a landmark-composed mask and a texture-restoration step keep edits local, and a DSPy program rewrites the user's request into a prompt the image model can act on. Before settling on the design reported here we tried the more obvious route, generating edits and judging them by eye, then attempting to replace that manual judgement with automatic image-quality metrics so the system could improve itself without a person in the loop. That attempt failed outright, the metrics did not agree with what our own annotators saw, and the failure is reported in full below because it is what forced the rest of the design, two blind paired experiments in which the source photo, mask, random seed, denoising strength and checkpoint are held constant so that the prompt is the only variable, judged by people rather than by a proxy.

The first experiment compared the rewriter against the hand-authored prompts of an earlier edit registry. 90 pairs rated for realism by six annotators, rewriting raised realism by 0.57 points on a five-point scale (Wilcoxon $p = 0.001$, $d = 0.43$), with the gain concentrated in the prompts that scored worst under the original wording. Because those baseline prompts were long and repetitive, we ran a second and harder experiment against what a user would actually type. Over 200 generated pairs and 240 blind judgements from twelve annotators, spanning 60 distinct requests across 15 edit types and 40 CelebA source faces, rewriting produced no overall benefit. Realism fell 0.08, identity 0.12, and edit compliance rose 0.17, none of the three surviving correction for multiplicity, and the rewrite was preferred in 47.6% of decided comparisons. Because 68 of the 172 unique Study 2 pairs received a second annotation, the 240 annotation tasks are not all independent image pairs; the confidence intervals are therefore more appropriate than a minimum-detectable-difference claim for describing the uncertainty.

The two results reconcile into a single account. Rewriting repairs badly written prompts but does not improve a well-formed request. One exception is large and systematic. For requests that ask to remove a discrete visual element such as makeup or facial hair, rewriting raised edit compliance by 1.31 points and was preferred 70.8% of the time, because the model handles negated visual concepts unreliably, so a prompt naming

the thing to be removed can still activate that visual concept. A differential prediction follows from that interpretation and is supported by the results. Removing a skin *condition*, where no such object concept exists, shows no benefit (-0.41 against $+0.45$ on realism, $p = 0.0017$).

We further report two constraints on how systems of this kind can be built. Eight automatic evaluators, including CLIP similarity, PickScore, the LAION aesthetic predictor and mask-cropped variants of each, failed to track human judgement (best $|\rho| = 0.181$), so these tested metrics were not considered suitable objectives for metric-driven prompt optimisation; the system therefore uses example-driven compilation instead. And from 480 human identity ratings, identity ratings are more strongly associated with how much of the face is regenerated ($\rho = -0.427$ with mask coverage) than with how strongly it is regenerated ($\rho = -0.148$ with denoising strength, with the within-band associations not reaching significance). Finally, we describe an experimental extension of the editing engine to a transformation that departs from every other edit in the system on purpose, rendering a subject as an infant, and report honestly where its realism currently falls short rather than overstating a feature with no annotated evaluation behind it yet.

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Chapter 1

Introduction

This chapter introduces the problem addressed by Vision Studio, the motivation for treating the wording of an edit request as a first-class engineering concern, the goals and contributions of the project, and the organisation of the remainder of the report.

1.1 Background and Motivation

Diffusion-based generative models have made photorealistic facial editing widely accessible, and cosmetic previews, showing a person a plausible "after" image for makeup, skin treatment or minor facial reshaping, are an increasingly common application. Two problems consistently limit how far such systems can be trusted.

The first is **identity drift**. When a model edits one attribute, for example softening wrinkles, it often alters unrelated identity cues such as bone structure or eye shape, so the person no longer recognisably looks like themselves [1, 2]. Generative cosmetic imagery carries a second problem alongside this. Wiegand et al. [3] evaluated four text-to-image generators by generating 9,060 patient images and having independent raters assess demographic characteristics. They found substantial demographic inaccuracies and an over-representation of White and normal-weight individuals. The finding concerns generation from prompts rather than editing of supplied photographs, so it does not directly establish bias in Vision Studio; it instead motivates caution about demographic representation in generative-image systems. Because cosmetic decisions are personal, a system that quietly changes who a person appears to be can cause real harm and erode trust.

The second problem is less often stated but, in our experience of building the system, at least as consequential. Diffusion models do not follow instructions. Their text encoder embeds every token as a visual concept, so the prompt is a description of a target image rather than a command. A request phrased naturally by a user is therefore not automatically a usable prompt, and two wordings of the same intention can produce visibly different results from an otherwise identical pipeline. The most striking case is negation. A prompt reading `remove his beard` contains the word `beard`, and the model duly draws one. Whether an automatic rewriting stage can close that gap, and where it can and

cannot, is the central question of this report.

Both questions are ultimately settled by human judgement, and reaching that conclusion took an attempt at avoiding it first. Early in development we generated a broad set of edits across many requests and inspected them directly, and a pattern was visible even before any formal scoring. Some results looked convincingly real and others carried a flaw that was easy for a person to see and hard to state as a rule, a slightly wrong skin tone at a mask boundary, an edit that was technically present but did not read as natural. The obvious next step, the one that would let the system judge its own output without a person in the loop, was to reach for an automatic image-quality metric. Recent progress in text-to-image generation has shown that collecting human preference ratings and training a reward model produces outputs better aligned with what people want [4–6], and general-purpose vision-language scoring is the standard tool for exactly this problem. We tried it and it did not work. The scores those models produced did not line up with what our own annotators saw when judging the same images blind, a result reported in full in Chapter 4. That failure is why the evaluation in this report rests on blind human comparison rather than an automatic score, and the failure itself, not just the workaround, is worth reporting, because it rules out an entire family of otherwise-reasonable optimisation strategies for this task.

1.2 Purpose and Goal of the Project

The goal of this project is to design and build a complete pipeline that (i) applies localized, identity-preserving cosmetic edits to face images from a free-text request, (ii) rewrites that request into a prompt suited to the diffusion backend, and (iii) measures, under controlled blind paired evaluation, whether the rewriting improves what people actually see. The contributions are:

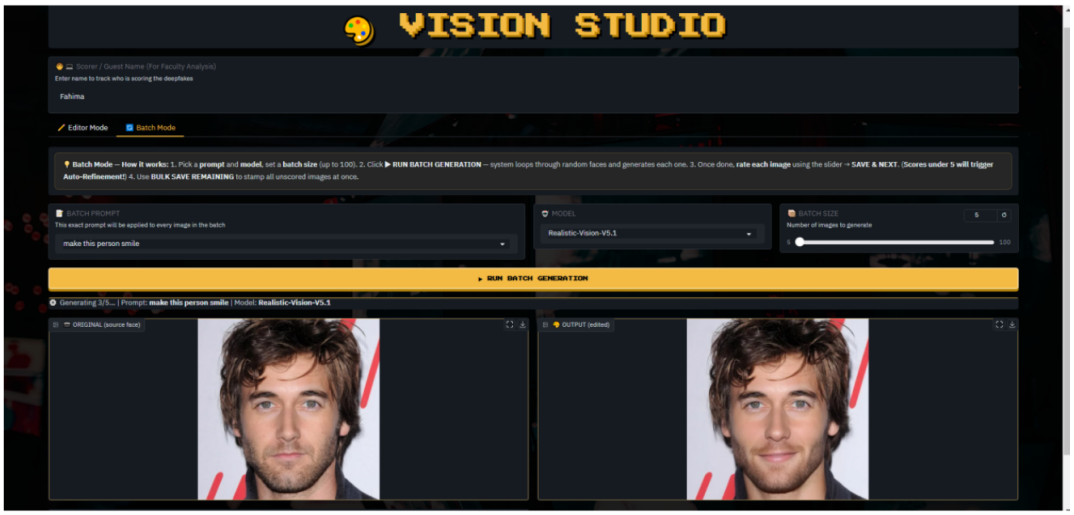
- **A composable region-masking scheme.** An edit resolves to a *set* of facial parts whose hulls are unioned, and a part is protected only if the edit does not target it. This replaces a fixed region enumeration that, because it unconditionally subtracted the eye and nose regions, silently produced empty masks for eye-makeup requests.
- **A DSPy prompt-rewriting stage with a deterministic contract check.** The task rules live in a declarative signature rather than a prompt string, and a validator checks each output for prose phrasing, hyperbolic vocabulary and negation contradictions before it reaches the generator.
- **Two controlled blind paired studies of that stage.** The first, against hand-authored prompts, measures a clear gain. The second, against the user's own plain-language request, measures none on average but isolates a large conditional effect for a specific, identifiable class of request. We report both together because the contrast between the two baselines is what explains the result.

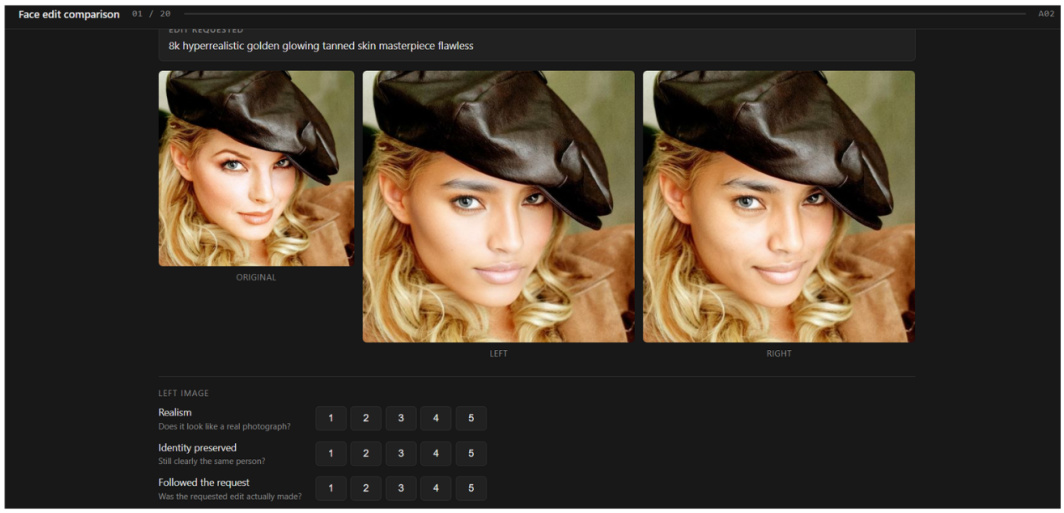
- **A mechanism, and a differential test of it.** We interpret the conditional effect as consistent with the difficulty of handling negated visual concepts, derive a prediction that distinguishes removal of a discrete element from removal of a skin condition, and test that prediction.
- **Two measured constraints on system design.** Eight automatic evaluators fail to track human judgement, so the tested metrics are not used as objectives for metric-driven DSPy optimisation, and identity ratings are more strongly associated with mask coverage than denoising strength in the evaluated dataset.
- **An experimental extension into a transformation identity preservation cannot cover.** Rendering a subject as an infant requires the opposite guarantee from every other edit in the system, so we describe it as a deliberate, narrow exception to the masking and prompting design, and report where its realism currently falls short rather than treating it as solved.
- **Reproducibility on commodity hardware.** The local diffusion and computer-vision components run on a single Kaggle GPU, while prompt rewriting uses a remote language-model backend. Every generated pair records the seed, prompts, mask specification and model settings, allowing an image to be regenerated under the same recorded configuration.

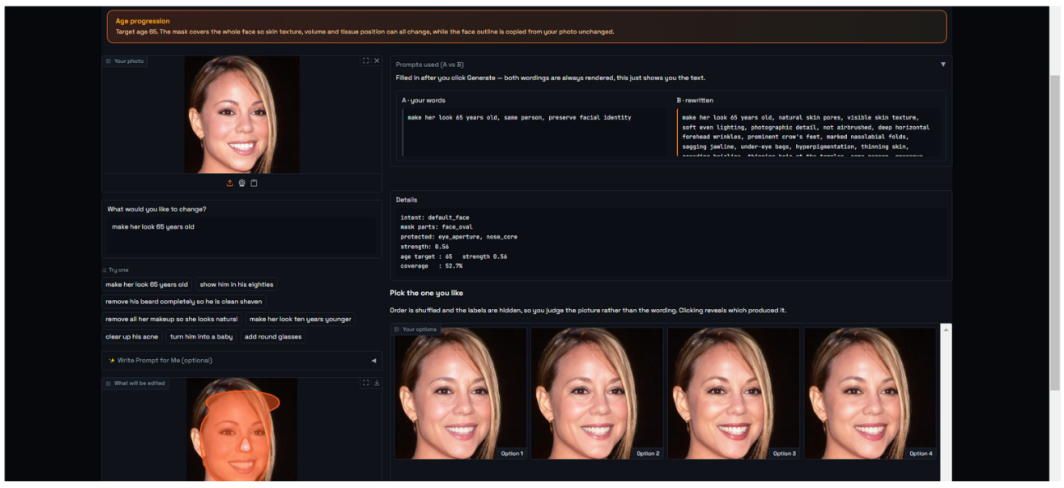
1.3 Evolution of Vision Studio: From Human Scores to an Evidence-Driven Editing System

Vision Studio did not begin as a fixed final system. The project evolved through several experimental stages in which the output of one stage influenced the design decisions of the next. This progression is important because several components of the final system, particularly the prompt-rewriting strategy and spatial masking direction, were selected from experimental evidence rather than being assumed to improve generation quality.

The project initially focused on a practical problem in diffusion-based facial editing: the same requested facial modification can produce substantially different results depending on how the request is written. At the same time, stronger or broader edits can unintentionally alter characteristics that make a face recognisable as the same person. Vision Studio was therefore developed as both an editing system and an experimental framework for investigating these two problems.







The figure shows the evolution of the user interface from the initial generation and human-scoring stage to the current identity-aware editing pipeline.

1.3.1 Stage 1: Initial Generation and Human-Scored Dataset

The first stage of the project established a reference dataset for studying the quality of diffusion-based facial edits. Source faces were edited across multiple requested transformations, including appearance changes such as ageing, facial-hair modification, makeup, and other local facial edits. The resulting images were not treated as self-validating outputs. Instead, human evaluators were used to assess perceptual properties that are difficult to define through a single automatic metric.

Human evaluation was important because facial editing involves several competing properties. An image can appear visually realistic while no longer resembling the original person. Conversely, an image may strongly preserve identity while failing to perform the requested edit. The project therefore treated perceptual quality as a multi-dimensional problem rather than reducing all outputs to a single similarity score.

The human-scored dataset served three purposes. First, it provided a reference against which automatic evaluators could later be tested. Second, it allowed comparisons between alternative prompt formulations under controlled generation settings. Third, it created a collection of successful and unsuccessful examples that could later be used as evidence for prompt-program compilation.

A key principle established during this stage was reproducibility. For each generated output, the project recorded the source image, edit request, generation settings, random seed, model information, and relevant mask configuration. Fixing the seed during paired comparisons is particularly important because diffusion generation begins from stochastic noise. If two systems are evaluated with different seeds, an observed difference may be caused by random variation rather than by the intervention being tested. By holding the seed fixed, paired outputs begin from comparable stochastic conditions.

This initial dataset therefore became the experimental foundation of the project rather than merely a collection of demonstration images.

1.3.2 Stage 2: Can Automatic Evaluation Replace Human Judgment?

After collecting human-scored facial edits, the next question was whether an automated vision-language evaluation pipeline could reproduce the same perceptual judgments. Automatic evaluation is attractive because human annotation is time-consuming and can be expensive at larger scale. If an automatic evaluator tracked human judgments reliably, it could potentially accelerate experimentation and provide an optimisation signal for the system.

The project therefore compared automatic scores with the previously collected human ratings. Rather than assuming that a modern vision-language model or image-quality metric represented perceptual truth, the automatic outputs were explicitly validated against human judgment.

The validation produced an important negative result: the evaluated automatic/VLM scores did not reliably reproduce the human realism judgments for the facial-editing do-

main. The relationship between the automatic scores and human scores was weak, with Pearson correlation $r = -0.065$ and Spearman correlation $\rho = -0.070$, while the mean absolute error was 1.084.

This result changed the direction of the project. Instead of using an automatic evaluator as the unquestioned objective function for optimisation, human judgments were retained as the perceptual reference. The negative result was also practically important for the DSPy component. Metric-driven prompt optimisation requires an evaluation signal that is meaningful for the target task. Optimising prompts against a metric that does not agree with human perception risks improving the metric without improving the generated image.

The failure of automatic evaluation was therefore not discarded as an unsuccessful experiment. It became a design constraint: human evidence would determine the experimental conclusions and would guide the selection of examples used in prompt optimisation.

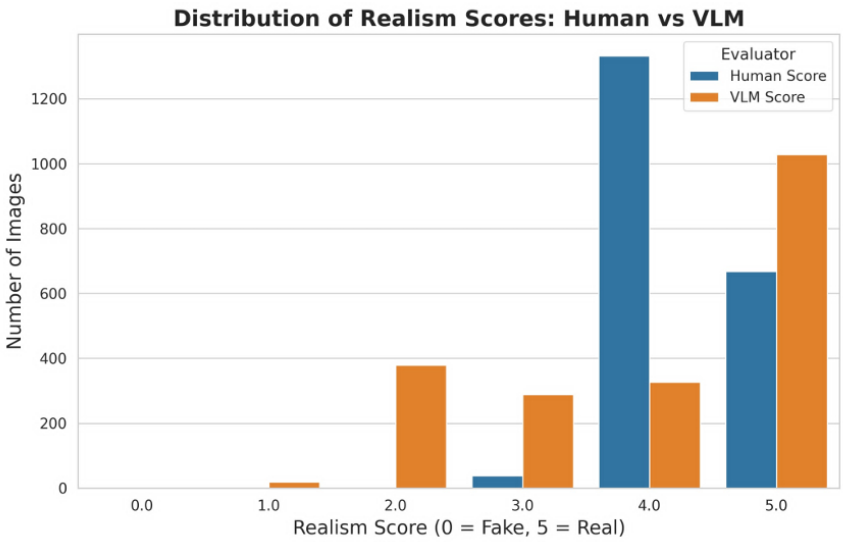


Figure 1.1: Comparison between automatic/VLM realism scores and human realism judgments. The weak correlation indicates that the evaluated automatic scoring approach did not reliably reproduce human perception in the masked facial-editing domain.

1.3.3 Stage 3: Expanded Human Evaluation

Because automatic evaluation did not provide a reliable substitute for perceptual judgment, the project expanded its human evaluation stage. A larger blind annotation study involving twelve annotators was conducted to compare generated outputs under controlled conditions.

The purpose of this stage was broader than simply obtaining another average score. Multiple annotators made it possible to investigate both agreement and variation in human judgments. Different evaluators do not necessarily assign identical scores to the same

image, particularly when judging subjective properties such as realism and instruction quality. For this reason, the evaluation protocol was designed to compare paired outputs while controlling the source image and generation configuration.

The blind design was important. Annotators were not informed which output was generated from the original user wording and which was generated from the rewritten prompt. This reduced expectation bias and allowed the comparison to focus on visible differences.

The resulting human judgments were used to investigate several questions: whether prompt rewriting consistently improved outputs, whether some types of requests benefited more than others, and which generation properties were associated with identity preservation.

This stage ultimately produced the evidence that shaped the next direction of Vision Studio. Prompt rewriting did not provide a uniform improvement across every request and evaluation dimension. At the same time, the analysis of identity preservation suggested that the spatial extent of editing was more strongly associated with identity change than denoising strength alone.

1.4 Organisation of the Report

Chapter 2 reviews related work on identity-preserving facial editing, human-preference reward modelling, and prompt optimisation, and positions Vision Studio against it. Chapter 3 details the methodology, covering the system design, the masking scheme, the DSPy rewriting stage and its algorithm, and the two evaluation protocols. Chapter 4 presents the results of both studies together with the metric-validation and identity findings. Chapter 5 discusses societal, ethical and environmental impacts. Chapter 6 presents planning and budget. Chapter 7 discusses the complex engineering problems and activities addressed. Chapter 8 concludes and outlines future work.

Chapter 2

Background and Literature Review

This chapter reviews the three research areas Vision Studio draws on. These are identity-preserving facial editing with diffusion models, human-preference reward modelling for generative images, and prompt optimisation as a programming discipline. Demographic bias in generative facial imagery cuts across all three and is treated where it bears on each. The chapter closes by explaining how our work differs from the most recent (2023–2026) contributions in each area.

2.1 Identity-Preserving Facial Editing

Latent diffusion models [7] underpin most modern facial editing. A central difficulty is editing a target attribute without disturbing identity, because age, expression and identity are entangled in the latent space. TimeMachine [1] performs fine-grained age editing by injecting age information through a multi-cross-attention module that separates age-related from identity-related features. IP-FaceDiff [2] and DiffFERV [8] address identity-preserving *video* editing through localized, inversion-based edits and preservation features. Instruction-following editors such as ACE [9] treat facial-attribute preservation as an explicit task. These methods achieve strong identity preservation but require substantial model training or fine-tuning and target generic or video edits.

Vision Studio instead uses a lightweight, training-free inpainting pipeline. Identity is protected structurally, through landmark-guided region masks, a conditional protection policy and a texture-restoration step, and it is checked at run time by a geometric guard that measures drift in bone-anchored landmarks (Section 3.1.4). We deliberately do not certify identity with a whole-image similarity score such as SSIM [10]. An inpainted composite is pixelwise identical to its source everywhere outside the mask, so a global similarity score is dominated by the region that cannot have changed, and Section 4.1 shows empirically that whole-image scoring of this kind does not track what annotators see. Identity is therefore reported against 480 human ratings rather than asserted through

a metric.

2.2 Spatially Controlled and Mask-Based Image Editing

A central design question in diffusion-based image editing is not only what semantic attribute should be changed, but also where the generative model should be permitted to modify the image. Mask-guided editing addresses this problem by restricting the region that participates directly in the editing process. This is particularly relevant for facial editing, where a requested modification may concern a localized feature while unnecessary regeneration of surrounding facial structure can introduce unintended changes.

Diffusion-based editing methods have explored spatial control through explicit masks, attention manipulation, inversion, and conditional generation. DiffEdit [11], for example, derives semantic editing masks from the difference between diffusion trajectories, while Blended Diffusion [12] combines text-guided generation with a spatial constraint that preserves image regions outside the target edit. Prompt-to-Prompt [13] demonstrates that attention maps can provide another mechanism for controlling which visual concepts are modified during editing.

More recent facial editing systems increasingly treat localization and identity preservation as related problems. Fine-grained editing requires sufficient spatial freedom to produce the requested transformation, while excessive modification can alter facial features that are important for recognisability. The resulting problem is therefore not simply one of minimizing the edited area. A very small region may preserve identity but fail to perform the requested transformation, whereas a broad region may perform the transformation successfully while unnecessarily regenerating identity-bearing structure.

This trade-off motivates the spatial design of Vision Studio. The system does not assume that a smaller mask is automatically better. Instead, landmark-guided facial regions are used to construct an edit space that can be aligned with the requested transformation. The project further investigates whether the spatial extent of permitted modification is associated with human-perceived identity preservation.

This distinction is important for the later stages of the project. The initial mask is not treated merely as an implementation detail. It becomes an experimentally meaningful variable that determines how much of the original face can potentially be regenerated. The current development direction therefore extends binary masking toward more structured spatial composition, including protected regions and regions with different editing priority.

Existing work provides powerful methods for spatial image editing, but the question of how spatial coverage relates to human-perceived identity preservation remains less directly characterised in lightweight, training-free facial inpainting pipelines. Vision Studio addresses this question experimentally by combining landmark-derived masks, controlled generation settings, and human identity ratings.

2.3 Human-Preference Reward Modelling

Aligning generative models with human preference has become a dominant theme. ImageReward [4] collected 137k human comparisons and trained a reward model that predicts a score for an image–prompt pair. PickScore, from the Pick-a-Pic dataset [5], and HPSv2 [6], trained on 798k preference choices, provide CLIP-based human-preference predictors, and Follow-Your-Preference [14] extends preference alignment specifically to image inpainting. These works establish that a learned reward model aligned to human ratings outperforms automatic metrics such as FID or Inception Score.

More recent work (2025–2026) continues to scale and refine these reward models. Examples include reward scaling for visual generation [15], dedicated image-editing reward models [16], personalized reward modelling [17], and personalized reward models that explicitly model individual taste [18]. Taken together they confirm that learned human-preference rewards remain the dominant alignment tool. They are, however, (i) general-purpose, trained on broad prompts rather than cosmetic edits, (ii) built from very large-scale annotation efforts, and (iii) not conditioned on the subject's demographics.

Vision Studio adopts the evaluative stance of this literature, that human preference is the quantity worth measuring, without adopting its instrument. We do not train a reward model. We collected a compact human-scored dataset for this domain and used it directly, both to answer the research question and, through `LabeledFewShot` demonstrations, to compile the rewriting program. The reason is reported in Section 4.1. When we tested the published rewards on our own annotated pairs, none of them tracked human judgement, which is a poor foundation on which to train a further reward model in the same domain.

2.4 Human Evaluation and Automatic Evaluation

Evaluation remains one of the central difficulties in generative image research. A generated image can be assessed according to multiple dimensions, including visual realism, similarity to a source subject, instruction following, aesthetic quality, and technical image quality. These dimensions are related but are not interchangeable. An image may, for example, appear realistic while failing to preserve the identity of the source subject, or it may preserve identity while failing to perform the requested edit.

Traditional image-quality metrics such as SSIM [10] and perceptual distances such as LPIPS [19] measure particular forms of image similarity but do not directly represent whether a human considers an edited face to remain the same person. Distribution-level metrics such as FID [20] are useful for comparing collections of generated images but are not designed to judge a single localized facial edit.

For this reason, human evaluation remains important in image generation and editing. Human annotators can simultaneously assess semantic and perceptual properties that are difficult to encode in a single automatic score. Preference datasets such as Pick-a-Pic and ImageReward demonstrate the value of collecting human comparisons, while learned

reward models attempt to approximate those judgments at scale.

However, the existence of a learned preference model does not imply that it transfers reliably to every visual domain. A reward model trained on broad text-to-image examples may encounter a distribution shift when evaluating localized facial inpainting. In such a setting, the evaluator must itself be validated before being used as a substitute for human judgment or as an optimisation objective.

This validation principle is particularly important for the present work. Vision Studio does not assume that an automatic score represents perceptual ground truth. Instead, the automatic evaluator is treated as a hypothesis: if it is to replace or guide human evaluation, its outputs must first demonstrate meaningful agreement with human ratings on the target task.

The distinction between automatic evaluation and human judgment directly shapes the methodology of the project. Human scores provide the reference against which automatic evaluators are tested, while the outcome of that validation determines whether a metric can honestly be used for downstream optimisation.

2.5 Personalization and Demographic Awareness

Very recent work argues that a single global reward model cannot capture individual or group taste. The 2026 personalized-taste dataset PAMELA [18] contributes dense per-image, multi-rater annotations *with user demographics*, arguing these are jointly required for personalized reward modelling. Surveys of personalized generation [21] similarly note that most personalization targets image *content* rather than aesthetic *preference*.

Our findings speak to this line indirectly. The present system does not condition on demographics, but the metric validation in Section 4.1 shows that the general-purpose reward models on which that literature rests do not transfer to masked facial inpainting at all. None of the published evaluators we tested correlated with our annotators' judgements. That is consistent with the argument for domain- and person-specific reward modelling, and it sets a concrete obstacle in front of it, since a reward model that does not track judgement in a domain cannot be personalised within it either.

2.6 Prompt Optimisation and Declarative Language Model Programming

Prompt quality can substantially influence the behaviour of modern generative models. In image generation, a user request may be incomplete, ambiguous, overly conversational, or expressed using language that does not clearly communicate the intended visual transformation. Prompt optimisation therefore seeks to improve the interaction between a user's intent and the language consumed by the underlying model.

Traditional prompt engineering typically treats the prompt as manually written text. Researchers or users modify wording, examples, instructions, or formatting and evaluate

the resulting model behaviour. This approach can be effective, but it becomes difficult to reproduce or systematically optimise when prompts are adjusted manually for each task.

More recent work treats prompting as a programmatic optimisation problem. Rather than viewing a prompt as a fixed string, a language-model pipeline can be represented through explicit input-output specifications, modules, demonstrations, and optimisation procedures. DSPy [22] is a prominent framework for this approach.

2.6.1 Prompt Optimisation for Generative Systems

A general prompt transformation can be represented as

$$p_{\text{out}} = f(r, c), \tag{2.1}$$

where r is the original user request, c represents additional task context, and p_{out} is the transformed prompt provided to a downstream model.

For generative image editing, the purpose of this transformation is not necessarily to make the prompt longer. Instead, the objective is to express the user's intended edit in a form that is sufficiently clear and specific for the image-generation model while preserving the original semantic intent.

The challenge is that prompt modification can introduce its own errors. A rewritten prompt may change the meaning of the request, introduce attributes not intended by the user, or over-specify visual details that interfere with identity preservation. Consequently, prompt optimisation should be evaluated according to the final generated output rather than according to the linguistic appearance of the rewritten text alone.

This is particularly relevant to facial editing. A request such as "make the person look older" may require a different spatial and semantic treatment from "add a beard." Prompt rewriting can clarify the intended transformation, but it cannot by itself determine whether the correct facial region is regenerated or whether the original identity remains recognisable.

2.6.2 DSPy as Declarative Language Model Programming

DSPy treats language-model interactions as declarative programs rather than collections of manually written prompts. A DSPy program specifies what information should enter a language-model module and what structured output is expected. The framework then provides mechanisms for compiling demonstrations and instructions into an executable language-model pipeline.

At a conceptual level, a DSPy program can be represented as

$$y = M_\phi(x; \theta), \tag{2.2}$$

where x is the structured input, M_ϕ is a language-model module, y is the structured

output, ϕ represents the module configuration, and θ represents the prompt-level parameters or demonstrations produced through compilation.

This separation is useful because it distinguishes the task definition from the particular wording used to communicate the task to a language model. A developer can therefore specify the required input and output fields while allowing the framework to select or compile supporting demonstrations.

In Vision Studio, the relevant task can be expressed conceptually as

$$(r, P) \rightarrow (p^+, p^-), \tag{2.3}$$

where r is the original user request, P represents relevant facial-region context, p^+ is the positive generation prompt, and p^- is an optional negative prompt.

The rewriting module is therefore not intended to replace the diffusion model or the spatial editing policy. It operates as an intermediate language transformation layer between the user's natural-language request and the image-generation backend.

2.6.3 DSPy Signatures and Structured Interfaces

A DSPy signature defines the expected inputs and outputs of a language-model operation. This provides a more structured interface than directly concatenating a large manually written prompt.

For a rewriting task, the signature can conceptually specify fields such as

$$\texttt{request}, \texttt{regions} \rightarrow \texttt{positive_prompt}, \texttt{negative_prompt}. \tag{2.4}$$

The structured interface has two advantages. First, it makes the intended information flow explicit. Second, it enables validation of the resulting output before it is passed to the diffusion model.

This distinction becomes important in an operational image-editing system. A language model may generate conversational text, explanations, or malformed output that is not appropriate for direct use as an image-generation prompt. By treating the output as a structured object, the system can validate required fields and apply deterministic repair or fallback procedures when necessary.

2.6.4 DSPy Modules and Compilation

DSPy programs can be composed from modules that perform particular language-model operations. A module may use demonstrations, reasoning procedures, retrieval, or other forms of context depending on the task.

Compilation refers to the process of configuring these modules using examples or evaluation signals. Conceptually, given a program $\mathcal{P}$ and a set of examples D, compilation produces a configured program

$$\mathcal{P}^* = \text{Compile}(\mathcal{P}, D, \mathcal{M}), \tag{2.5}$$

where $\mathcal{M}$ represents an optional evaluation mechanism.

Different compilation strategies make different assumptions about the availability of such an evaluation mechanism. Some optimisers require a metric that can score candidate outputs. Others can construct few-shot programs from labelled demonstrations.

This distinction is central to Vision Studio because the choice of DSPy optimisation strategy depends directly on whether an automatic metric can be trusted in the target domain.

2.6.5 DSPy Optimisers and Demonstration Selection

DSPy provides multiple approaches for improving language-model programs. Broadly, these approaches differ according to how candidate instructions and demonstrations are selected.

Metric-driven optimisation methods search for prompt configurations that improve a defined evaluation function. If a reliable metric exists, the optimisation problem can be expressed as

$$\theta^* = \arg\max_{\theta} \mathbb{E}_{(x,y)} \left[m \left(M_\theta(x), y \right) \right], \tag{2.6}$$

where m is an evaluation metric and θ represents the prompt configuration or demonstration set.

This approach is attractive when the metric accurately represents the desired outcome. However, optimisation is only as meaningful as the evaluation signal. If a metric does not agree with human judgment in the target domain, maximising that metric can improve the score while failing to improve the actual output quality.

Example-driven approaches instead use labelled demonstrations directly. A set of examples

$$D = \{(x_i, y_i)\}_{i=1}^{N} \tag{2.7}$$

can be incorporated into the language-model program as demonstrations of the desired input-output behaviour.

`LabeledFewShot` follows this latter principle. It constructs a few-shot configuration from labelled examples without requiring a separate learned reward function to rank every candidate output. This makes example-driven compilation particularly relevant when reliable labelled examples exist but a validated automatic metric does not.

The choice between metric-driven and example-driven optimisation should therefore not be viewed simply as a software preference. It is an empirical decision that depends on the validity of the available evaluation signal.

2.6.6 Human-Guided Example-Driven Compilation

The relationship between human evaluation and DSPy is particularly important in Vision Studio. The project began with human-scored facial edits and subsequently investigated whether automatic evaluators could reproduce those judgments. This evaluation step was necessary before selecting an optimisation strategy.

If an automatic evaluator reliably tracked human perception, it could potentially serve as the metric m in a metric-driven prompt optimisation loop. Candidate prompt transformations could then be generated and selected according to their expected score.

However, if automatic scores fail to agree with human judgment, the same optimisation strategy becomes difficult to justify. In that situation, improving the automatic metric does not necessarily imply improving realism, identity preservation, or instruction following as perceived by humans.

Vision Studio therefore uses human evidence in a different way. Rather than training a new reward model from the compact dataset, selected labelled examples are used as demonstrations for example-driven compilation. The human-scored dataset consequently plays two roles: it provides the perceptual reference for evaluation, and it provides evidence for selecting useful prompt demonstrations.

This creates an evidence-driven pipeline:

$$\textit{Human annotations} \rightarrow \textit{validation of automatic evaluation} \rightarrow \textit{selection of optimisation strategy} \rightarrow \textit{example-driven DSPy compilation.}$$

The importance of this pipeline is methodological. DSPy is not assumed to improve the system simply because it is used. Its optimisation strategy is constrained by the evaluation evidence collected within the project.

2.6.7 Limits of Prompt Rewriting for Facial Editing

Prompt optimisation operates in language space, whereas many failures in facial editing occur in image space. A rewritten prompt may express the requested transformation more clearly, but the diffusion model must still determine which facial structure to modify and how broadly to regenerate the target image.

This creates a fundamental limitation: improved wording does not necessarily guarantee improved identity preservation. Some requests require modification of broad facial regions, while others can be performed locally. Consequently, prompt rewriting and spatial control address different parts of the editing problem.

This distinction motivates the broader architecture of Vision Studio. DSPy is used to improve the expression of user intent, while landmark-guided masking constrains where the diffusion model is permitted to implement that intent. Human evaluation is then used to assess whether these interventions improve the resulting image.

The project therefore treats prompt optimisation as one component of a larger editing pipeline rather than as a complete solution to facial image quality.

2.7 Positioning of This Work

Relative to identity-preserving editors [1, 2, 8], Vision Studio protects identity structurally, through composed masks and texture restoration, rather than by retraining a backbone, and it measures the result against human identity ratings instead of asserting it through a similarity metric. Relative to preference reward models [4–6, 14], our contribution is negative and, we think, useful. We test several of them directly on this task, including on the edited region alone, and find that none tracks human judgement, which constrains which optimisation strategies are honestly available.

Prompt optimisation provides the third component of the research positioning. As reviewed in Section 2.6, DSPy [22] treats language-model interactions as compilable programs rather than fixed manually written prompts. Our contribution is not the introduction of DSPy itself, but the empirical decision about how it can be used in this domain. Because automatic evaluators did not reliably track our human reference judgments, metric-driven prompt optimisation was not considered a valid optimisation strategy. The project therefore uses human-guided labelled examples with an example-driven compilation approach. This connects the evaluation findings directly to the architecture of the rewriting system.

Chapter 3

Methodology

This chapter describes the design of Vision Studio, the software components used to implement it, the editing pipeline, the DSPy prompt-rewriting stage and its algorithm, and the two evaluation protocols under which the rewriting was tested.

3.1 Overview of Final System

Vision Studio is organised as the pipeline shown in Figure 3.1. A source photograph and a free-text edit request enter the system. The request is interpreted to determine the facial regions relevant to the requested edit, from which a landmark-guided mask is constructed. In parallel, the DSPy component rewrites the request into a diffusion-oriented prompt.

For controlled evaluation, two images are rendered under the same source image, mask, random seed, denoising strength and generation configuration. The two arms differ only in the prompt formulation. This allows the effect of prompt rewriting to be evaluated without confusing it with randomness in diffusion generation.

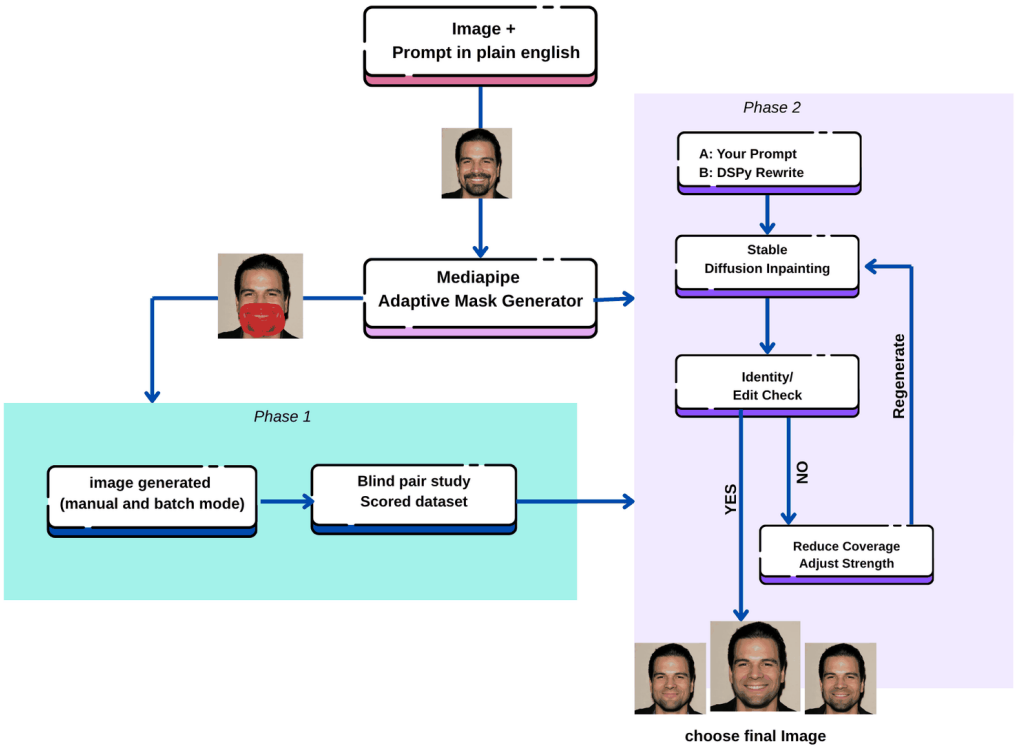


Figure 3.1: Vision Studio system architecture. The dashed band marks the variables held constant across the two arms; only the positive prompt differs, so any measured difference between arms is attributable to the rewriting.

3.1.1 Editing Engine

Given a face image and a mask specification, the engine locates 468 facial landmarks with MediaPipe FaceMesh [23] and constructs a binary edit mask. A latent diffusion inpainting model [7] regenerates only the masked region under the positive prompt; everything outside the mask is preserved by an alpha composite against the original, which is why every pixel outside the resolved mask is preserved by alpha compositing; hair, clothing and background are therefore unchanged when they fall outside that mask. A final *texture-restoration* step re-injects the high-frequency detail of the original, pores and sheen, into the edited region at 30% strength. Without it, inpainted skin acquires the smooth plastic appearance characteristic of diffusion output. Generation is performed at the model's native 512×512 resolution and resized back to the input aspect ratio.

3.1.2 Composable Region Masking

The masking scheme is the component that changed most between the two studies, and the reason is worth recording. In the earlier design an edit was mapped to exactly one of eight fixed regions, and the mask builder then unconditionally subtracted a protection region covering the eyes and the nose, on the reasoning that those areas carry identity. The consequence was that an eye-makeup request resolved to the `face` region and then

had the eyelids removed from its own mask, so the edit had nowhere to act and silently did nothing.

In the present design an edit resolves to a *set* of parts whose convex hulls are unioned, and protection is *conditional*: a part is subtracted only if the edit does not target it. Eye makeup keeps the eyelids and the brow and protects only the eye aperture, so eye colour and gaze are never regenerated while the lid above them remains editable. Full-face makeup covers the whole face oval including the nose, because contouring and highlighting require the nose bridge. Multiple intents compose, so a request for a smokey eye and a red lip masks eyelids, brows and lips in a single pass. Figure 3.2 shows the resulting masks and their coverage.

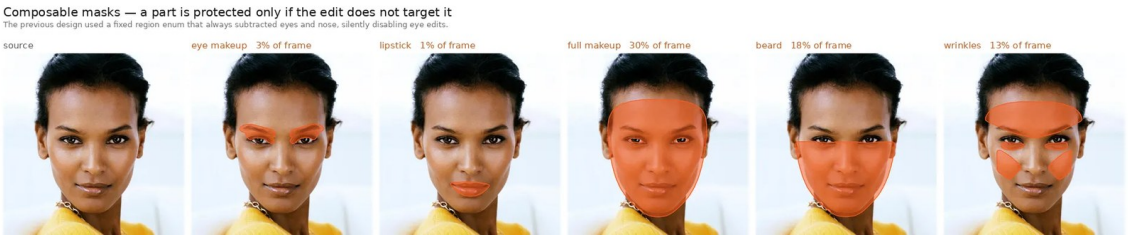


Figure 3.2: Composable masks for five edit intents on one source face, with the fraction of the frame each covers. Orange marks the editable region. The previous fixed-region design always subtracted the eye and nose areas, which is why eye-makeup edits produced no visible change.

Each specification carries its own growth factor, feather radius and denoising strength, and the resolved specification, including the part list, the protected list and the measured coverage, is written into the dataset alongside every generated image. Masking is therefore a recorded variable rather than a hidden one, which is what makes the coverage analysis in Section 4.7 possible.

3.1.3 Free-Text Intent Parsing

The earlier system operated over a fixed registry of hand-authored edits, which meant the space of possible edits was bounded by what the designers had written down. The present system parses free text. Each intent is matched by a word-boundary regular expression and contributes a part set, a protection set, a growth factor and a strength; where several intents match, parts are unioned, protection sets are unioned and then reduced by any part that some matched intent targets, and strength is the maximum over matched intents. Requests that fall outside the reachable area, hair, clothing, background, eyewear, are detected and reported to the user rather than silently producing an unchanged image.

3.1.4 Identity Guard and Risk Reporting

Masking bounds *where* the image can change, but it does not bound how far the regenerated region drifts from the person inside it. A geometric guard therefore runs after

generation. It recovers the FaceMesh landmarks from the output and compares the bone-anchored subset, the outer eye corners, the nasion, the nose base and the jaw outline, against the same landmarks on the source, after similarity alignment. These are the points that move least with expression and most with identity. If the residual exceeds a threshold the image is discarded and regenerated with the coverage reduced and the denoising strength lowered, which is the feedback path drawn on the right of Figure 3.1. Two attempts are allowed before the system returns the best of what it has.

The guard is a run-time safeguard, not the evidence for identity preservation in this report. It tests a geometric proxy that has never itself been validated against human perception, which is exactly the kind of claim Section 4.1 shows should not be trusted without checking. The identity claims in Chapter 4 rest on 480 human ratings instead.

Alongside the guard the system reports an identity-risk band derived from measured mask coverage, using the relationship established in Section 4.7. A request resolving to a mask over roughly a third of the frame is labelled high risk before it is rendered. The point is to state the trade-off rather than resolve it silently, because Section 4.7 also shows that lowering the strength does not buy the identity back

3.1.5 Data Layer

Every generated pair is recorded as one row of a flat CSV file together with the four images it produced: the source, the mask, and the two rendered arms. The row stores both positive prompts, the fixed rendering negative prompt, any rewriter-proposed negative prompt and the provenance of the rewrite, the full mask specification, the random seed, the denoising strength, and the checkpoint, guidance scale, step count and resolution. Because the seed and both prompts are recorded, any image in the dataset can be regenerated exactly from the row alone; this was used in practice to reconstruct the image set for the metric-validation experiment after a session reset, without access to the original files.

3.1.6 Controlled A/B Rendering

The prompt-rewriting experiment was implemented as a controlled paired comparison. For every source image and edit request, two outputs were generated. Arm A used the original prompt formulation, while Arm B used the DSPy-generated rewrite.

All generation variables other than the positive prompt were held constant. These included the source image, edit mask, random seed, denoising strength, checkpoint, guidance scale, number of inference steps, resolution and the negative prompt.

Let the generation process be represented as

$$I = G(x, m, p, z, s, \theta), \tag{3.1}$$

where x is the source image, m is the mask, p is the prompt, z is the random seed, s is the denoising strength and θ represents the remaining generation settings.

3.1.7 Deployed A/B Behaviour

The controlled A/B structure used for evaluation can also be retained in the deployed interface. The user's original wording and the DSPy rewrite are rendered under the same generation settings and presented in randomised order. The user may select the preferred result rather than having the rewritten prompt silently replace the original request. When enabled in the deployed interface, the selected result and generation metadata are recorded for later analysis.

The two experimental arms are therefore

$$I_A = G(x, m, p_A, z, s, \theta), \tag{3.2}$$

$$I_B = G(x, m, p_B, z, s, \theta). \tag{3.3}$$

Thus, the prompt is the intended difference between the two generated outputs, while all other variables are held constant. Fixing the random seed is particularly important because diffusion models otherwise produce different sampling outcomes even under the same settings.

3.2 DSPy Prompt Rewriting

Prompt rewriting was introduced because users can express the same edit request in very different forms, including short phrases, conversational sentences, grammatically incomplete requests and already well-structured prompt tags. The purpose of the DSPy component was not to assume that rewriting automatically improves image generation, but to systematically test whether transforming user wording affects human-rated realism, identity preservation and instruction following.

3.2.1 Why a Program Rather Than a Prompt String

The direct approach to prompt construction is string interpolation, for example `f"a photo of a face, {request}, high quality"`. This couples three separate concerns into one literal: the task rules, the output format, and the user's input. Changing a rule means editing a string, nothing verifies that the model complied, and the resulting prompt cannot be improved from data without hand-editing.

DSPy [22] separates these. A *Signature* declares the input fields, the output fields, and the task rules; a *Module* selects the reasoning strategy; and an *Adapter* renders the signature into the message actually sent to the language model and parses the reply back into typed fields. No monolithic prompt template is used for the rewriting task. Because the task is declared through a signature rather than a single prompt literal, it can later be compiled against labelled data, as described in Section 3.2.6.

The complete rewriting component is provided as a standalone documented source

file, `vision_studio_dspy.py`, accompanying this report. The description below covers its structure; the file itself carries the implementation and the reasoning behind each rule.

3.2.2 Signature

The signature takes the verbatim user request and the list of facial parts the mask permits, and returns a positive and a negative prompt. Its docstring carries the task rules, each of which is traceable to an observation from our own experiments:

1. **Comma-separated visual tags only, no sentences and no instruction verbs.** The text encoder embeds words such as `ensure`, `while` and `maintain` as visual concepts, so they contribute noise rather than direction.

2. **No hyperbolic quality vocabulary** (`4k`, `8k`, `hyperrealistic`, `masterpiece`, `RAW photo`). In Study 1 the one prompt written in that register, `enhance to hyperrealistic 4k resolution`, drew the strongest single gain from rewriting once the hyperbole was stripped (Table 4.3).

3. **Preserve intent; for a removal request, never name the thing removed.** Describe the end state instead: `remove beard` becomes `clean shaven jaw, bare skin, faint shadow`. This is the rule that carries the effect measured in Section 4.6.

4. **Describe only what lies inside the masked region.** Tokens describing hair, clothing or background cannot be rendered and only dilute the embedding.

5. **Six to twelve tags, under forty words.**

3.2.3 Module and Validator

The module is `ChainOfThought` [24], which extends the signature with a reasoning field so the model works through the transformation before emitting tags. This supports rule 3: the model must first determine what the face will look like once the element is gone, and only then describe that state.

DSPy guarantees the *structure* of the reply but not that its *content* obeys the rules, and a lite-tier model frequently does not. A deterministic validator therefore checks each output for hyperbolic vocabulary, instruction verbs, sentence punctuation, insufficient tag count, and, for removal requests, whether the prompt still names the attribute being removed. Hyperbole and instruction verbs are additive noise and can be deleted safely, so those two violations are repaired and the output re-checked; anything else is flagged and recorded, so a defective rewrite remains visible in the dataset rather than being hidden. Every row carries the provenance of its rewrite, so rows where the language model was unreachable can be excluded from any claim about the rewriter.

3.2.4 Language Model Backend

DSPy ordinarily reaches a hosted model through LiteLLM. That path failed here for two independent reasons: LiteLLM's model registry rejected the current Gemini model identifiers, and the older identifiers it did accept had no remaining free-tier quota. Rather than work around the abstraction we implemented `dspy.BaseLM` directly against the REST endpoint. The contract is narrow, `forward()` must return an object shaped like a chat completion, because that shape is what the adapter parses, so roughly forty lines of code are sufficient to place any model behind a DSPy program. All prompts in Study 2 were produced by `gemini-3.1-flash-lite` through this backend, and the backend identifier is recorded on every row.

A practical detail worth documenting: `dspy.configure()` binds a language model to the calling thread. The editing interface serves each request in a worker thread, where the global binding is absent, so every call raised and fell back silently. Holding the model object and binding it per call inside `dspy.context()` is the fix, and Algorithm 3.1 reflects it.

Algorithm 3.1: Prompt rewriting with contract validation

Input: user request r; mask part list P; language model $\mathcal{L}$; cache C

Output: positive prompt p^+, negative prompt p^-, provenance s, flag list F

if r *is empty* **then return** $(\varnothing, \varnothing, \textsc{empty}, \{\text{empty}\})$;
$k \leftarrow (r, P)$;
if $k \in C$ **then return** $C[k]$ with $s \leftarrow s \,\|\, \textsc{cache}$;
$s \leftarrow \textsc{fallback}$; $F \leftarrow \varnothing$; $p^+ \leftarrow \varnothing$;
begin try
    `// bind the LM to this thread, then run the compiled module`
    with $\textsc{context}(\mathcal{L})$ **do**
     $y \leftarrow \textsc{ChainOfThought}(\textsc{RewriteEditPrompt})(r, P)$;
    $p^+ \leftarrow \textsc{normalise}(y.\mathit{positive})$; $p^- \leftarrow \textsc{normalise}(y.\mathit{negative})$;
    $F \leftarrow \textsc{validate}(p^+, r)$;
    if $F \neq \varnothing$ ***and*** $F \subseteq \{\mathit{HYPE}, \mathit{PROSE}\}$ **then**
        $p^+ \leftarrow \textsc{repair}(p^+)$; $F \leftarrow \textsc{validate}(p^+, r)$; $s \leftarrow \textsc{repaired}$;
    else if $F = \varnothing$ **then**
        $s \leftarrow \textsc{dspy}$
    else $s \leftarrow \textsc{flagged}$;
begin catch e
    $F \leftarrow \{\textsc{exception}(e)\}$;
if $p^+ = \varnothing$ **then**
    `// deterministic fallback; such rows are excluded from claims about the rewriter`
    $p^+ \leftarrow \textsc{normalise}(r) \,\|\, \textsc{TextureTags}$; $s \leftarrow \textsc{fallback}$;
if $p^- = \varnothing$ **then** $p^- \leftarrow \textsc{BaseNegative}$;
`// a failure is never cached, so one unreachable-model error cannot replay`
if s *begins with* $\textsc{dspy}$ **then** $C[k] \leftarrow (p^+, p^-, s, F)$;
return (p^+, p^-, s, F);

The final line is not incidental. An earlier revision cached every result including failures, so a single unreachable-model error was replayed for 140 subsequent rows before it was noticed. Caching only successful rewrites makes the failure mode self-limiting.

3.2.5 Prompt Assembly and the Identity Anchor

Both arms receive the same identity anchor, `same person, preserve facial identity`, appended to the positive prompt. The anchor belongs to the rendering harness rather than to the intervention under test, so it must be identical on both sides or the comparison is confounded. Arm A is the user's request plus the anchor; arm B is the rewrite plus the same anchor. The negative prompt is fixed and shared between both arms for the

controlled comparison. The rewriter may return a negative-prompt field for logging and future use, but that field is not used to distinguish the two arms.

3.2.6 Optimisation from Human Preferences

DSPy offers three optimiser families. `MIPROv2` searches over instructions and demonstrations against a metric; `BootstrapFewShot` generates candidate demonstrations and keeps those a metric accepts; `LabeledFewShot` selects demonstrations from labelled examples and requires no metric at all.

The first two were not considered methodologically justified here because they require a reliable metric for scoring candidate outputs, and no such metric was validated for this task. As Section 4.1 reports, the eight tested automatic evaluators all showed weak agreement with human judgement, the strongest absolute Spearman correlation being 0.181. Driving an optimiser with one of those unvalidated scores could therefore optimise the evaluator rather than the intended perceptual outcome. `LabeledFewShot` needs only labelled examples. We constructed such examples from human-preferred rewrites in Study 2; LabeledFewShot then selects demonstrations from that labelled set, restricted to the request type for which the compiled module is used. The Study 2 results reported in Chapter 4 are the evaluation of the pre-compilation rewriter; the human-preferred examples are used only for the subsequent example-driven compilation step, not as the metric for the same evaluation.

3.3 Human Evaluation Studies

The final quality of a facial edit was evaluated primarily through human judgement. Automatic image evaluators and vision-language models were initially investigated as possible evaluation tools; however, as reported in Section 4.1, they did not reliably track human judgement in this masked facial editing domain.

Human evaluation was therefore used as the primary source of evidence. Two controlled studies were conducted. Study 1 compared DSPy rewriting against the hand-authored prompts used in an earlier version of the system. Study 2 used a more realistic baseline by comparing the rewrite directly against the wording a user would naturally provide.

3.3.1 Study 1: Against Hand-Authored Prompts

The first study evaluated the rewriter against the hand-authored positive prompts of the earlier edit registry. For each source face, arm A was rendered from the registry prompt and arm B from the same specification with only the rewritten positive prompt substituted. Ninety pairs were scored blind on realism by six annotators in the same session, so leniency and ordering effects cancel within a pair. Scores were collected specifically for this comparison rather than reused from an earlier non-blind batch scoring, which had shown a wide annotator-leniency spread.

Both authors were among the six annotators. Scoring was blind with respect to which side carried the rewrite, but author participation remains a potential source of evaluator bias and is therefore reported as a limitation. Study 2 was annotated by twelve people, none of them authors, and the negation finding on which this report's central claim rests comes from that study rather than this one.

3.3.2 Study 2: Against the User's Own Wording

The registry prompts of Study 1 were long and internally repetitive. Several stated `same gender` twice and `visible pores` twice within a single prompt, and Table 4.3 reproduces them in full. A rewriter that improves on a baseline like that has demonstrated something narrower than the claim we wanted to make. Study 2 therefore replaces the baseline with what a user would actually type, and nothing else about the protocol is relaxed. If anything it is tightened, with twice as many annotators, more than twice as many pairs, three rated dimensions instead of one, and the source photograph visible throughout so that identity can be judged against a reference.

Sixty distinct requests were written across fifteen edit categories, each category expressed at four registers, in order to span the range of input quality a deployed system would receive:

- **bare**, one or two words, e.g. `lipstick`;
- **prose**, a conversational sentence, e.g. `can you give her a smokey eye for a wedding`;
- **hype**, loaded with quality vocabulary, e.g. `8k hyperrealistic masterpiece flawless makeup`;
- **tagged**, an already well-formed comma-separated tag list, serving as the control condition in which no improvement is expected.

Two hundred pairs were generated across 40 CelebA source faces, with the source photograph, mask, random seed, denoising strength, negative prompt, checkpoint (Realistic Vision V5.1 inpainting), guidance scale (7.0), step count (28) and resolution (512) held constant within each pair.

The annotation budget was twelve annotators judging twenty pairs each, giving 240 judgements, and 68 pairs were deliberately assigned to a second annotator so that agreement could be measured. Those 240 judgements therefore cover 172 distinct pairs of the 200 generated; the remaining 28 were not reached and are excluded from every analysis in Chapter 4. None of the twelve annotators was an author of this report. Left and right position was randomised per pair, the source photograph was displayed throughout so that identity could be assessed against a reference, and the annotator was never told which side was which. Each side was scored on realism, identity preservation and edit compliance, and a forced-choice preference was recorded for the pair.

Per-annotator mean ratings spanned 2.07 points, so scores were centred within annotator before aggregation. All paired contrasts use the Wilcoxon signed-rank test, preference proportions use an exact binomial test, and families of related tests are Holm-corrected. Analyses declared exploratory in Chapter 4 are reported uncorrected and labelled as such.

3.3.3 Annotation Workflow

Each annotation task displayed the source photograph together with two generated outputs. The left–right position of the outputs was randomised so that annotators could not determine which image was generated from the original prompt and which was generated from the DSPy rewrite.

Annotators evaluated realism, identity preservation and instruction following separately. A forced-choice preference between the two outputs was also collected. Evaluating these dimensions separately was important because an edit could successfully follow the requested instruction while appearing less realistic or while changing the perceived identity of the subject.

3.3.4 Statistical Analysis

Human ratings were analysed as paired observations because the two images within each comparison were generated from the same source image and edit request. Differences between the two experimental conditions were tested using the Wilcoxon signed-rank test.

Forced-choice preferences were analysed using an exact binomial test. Where multiple related comparisons were performed, Holm correction was applied [25]. Relationships between variables such as mask coverage, identity preservation and geometric drift were analysed using Spearman's rank correlation coefficient [26].

Because annotators differed substantially in scoring strictness, ratings were centred within annotator before aggregation. Analyses identified as exploratory are reported separately and interpreted accordingly.

3.4 Boundary Case: Extreme Age Transformation

The following case is included as a boundary condition rather than as part of the controlled A/B prompt-rewriting studies. It illustrates a situation in which the two central objectives of Vision Studio—preserving identity and following the requested transformation—become fundamentally conflicting. Every mechanism described so far protects identity by construction: a part is masked only where the edit requires it, and the parts most associated with recognising a person, the eyes and the bridge of the nose, are withheld from editing whenever the request does not specifically target them. Ageing a subject by forty years still leaves that structure in place, because the underlying bone geometry of an adult face changes only slightly with age. A request to render the same subject as an infant cannot

honour that guarantee, because an infant's proportions genuinely are different, a proportionally larger cranium relative to the face, a flatter and smaller nose bridge, fuller cheeks, and a face that has not yet grown into the bone structure the identity protection elsewhere in this system exists to preserve. Treating this as an ordinary edit and applying the standard protection set would leave the output looking like an adult face with infant colouring rather than an infant face, which defeats the purpose of asking for the transformation at all.

The system therefore treats an infancy request as a deliberate, narrow exception to the design described in Section 3.1.2. The nose is left unprotected, since its shape is one of the clearest signals of the requested transformation, while the eye region keeps the same hard floor applied everywhere else in the system: the iris and sclera are never regenerated, whatever the request. The identity guard used elsewhere, which measures drift in bone-anchored landmarks and regenerates the image if it exceeds a threshold, is bypassed specifically for this request type, because its purpose is to catch unintended structural drift, and here the structural change is the entire point; applying it would simply cause the guard to fight the transformation on every attempt.

One further adjustment was needed after the first attempts at this feature looked wrong in a specific way: the output resembled an adult face with an infant's colouring and skin rather than a coherent infant face. The cause was in the prompt rather than the mask. Every other request in the system appends a fixed identity-anchoring phrase, `same person, preserve facial identity`, to keep the two rendered arms comparable and to discourage unwanted drift. Sent alongside a request for infant proportions and the absence of visible bone structure, that phrase directly contradicts the rest of the prompt, and the model produced a compromise between the two instructions rather than a committed result. The fix was to give this request its own anchor, one that asks for a photorealistic result without asserting that identity is to be preserved, since for this transformation it is not.

This feature has not yet been evaluated under the blind paired protocol used for the rest of this report, and the discussion of its current limitations is deferred to Section 4.8.1.

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3.5 Implementation Environment

Source faces are drawn from CelebA [27]. The local diffusion and computer-vision components run on a single Kaggle T4 GPU; the reported mean generation time is 6.5 seconds per image. Prompt rewriting uses the Gemini 3.1 Flash Lite REST backend described above. Table 3.1 lists the principal tools and the reason each was selected.

Table 3.1: List of Software Tools

Tool	**Function**	**Similar Tools**	**Why Selected**
Stable Diffusion (inpainting)	Localized face editing	SDXL, FLUX	Strong quality at low VRAM; mask-based control
MediaPipe FaceMesh	Landmarks and composable region masks	dlib, Face++	Accurate, fast, license-friendly, on-device
DSPy	Declarative prompt programming and few-shot compilation	LangChain, raw API calls	Task rules held as a signature rather than a string, so the program can be compiled from labelled data
Gemini 3.1 Flash Lite	Language model behind the rewriter	GPT-4o-mini, local Qwen2.5	Free tier was adequate during the study period; reached through a custom REST backend
CSV + PNG archive	Dataset storage	SQLite, Supabase	Portable, diff-able, and directly readable by the analysis notebooks
SciPy / scikit-image	Statistical testing and image analysis	R, statsmodels	Non-parametric paired tests, correlation and mask geometry in one environment
Gradio	Editing and demonstration interface	Streamlit	Rapid UI with shareable links

Chapter 4

Results and Discussion

This chapter presents the experimental findings of Vision Studio in the order in which the evaluation strategy developed. The project initially investigated whether automatic vision-language and image-quality evaluators could provide a scalable alternative to human judgement. Their failure to track the collected human ratings led to human evaluation becoming the primary source of evidence.

The chapter then reports two controlled studies of DSPy prompt rewriting. Study 1 compares rewriting against the hand-authored prompts used in the earlier system, while Study 2 uses a more realistic baseline consisting of users' own free-text requests. The results are followed by an analysis of when rewriting helps, the relationship between spatial mask coverage and identity preservation, and the limitations revealed by an extreme age transformation.

4.1 Validation of Automatic Evaluators

Before relying on large-scale human annotation, the project investigated whether automatic evaluators could approximate human judgement for masked facial editing. Earlier experiments included LLaVA-based scoring and perceptual similarity measures, while the final evaluation compared several published image-text, preference and aesthetic evaluators against the collected human judgements.

The motivation was practical. If an automatic evaluator reliably ranked edited outputs in the same direction as human annotators, it could serve both as a scalable quality measure and as an optimisation objective for automated prompt improvement. The following experiment therefore tested this assumption directly rather than treating a published metric as valid by default. A metric that predicted human judgement would allow prompt optimisation to run without annotators, and could provide an evaluation signal for metric-driven DSPy optimisers. We therefore scored every annotated image with published evaluators and correlated each against the human ratings, using paired differences (Δmetric against Δhuman) rather than raw scores, since that is the quantity an optimiser would have to rank. To keep the comparison fair, both arms were scored against the *same*

text, the user's original request, rather than each against its own prompt: scoring arm B against its own longer and more visually loaded rewrite would inflate it by construction.

None of the eight evaluators tracked human judgement (Figure 4.1). CLIP similarity [28] reached $\rho = -0.181$, the strongest of the eight in absolute terms and still far below anything usable; PickScore reached -0.093 and the LAION aesthetic predictor -0.014.Together with the earlier Vision-Language Model experiment using LLaVA-1.5-7B, which produced a Pearson correlation of $r = -0.065$ and a Spearman correlation of $\rho = -0.070$ against human realism ratings, the tested automatic evaluation approaches consistently failed to track human judgement.

Because inpainted composites are identical to the source outside the mask, and identical between the two arms there, a global score is computed largely over pixels that cannot differ. Mask coverage averages roughly 20% of the frame, so this is the most plausible explanation for the null, and we tested it: all three image-based evaluators were recomputed on the mask bounding box alone. Cropping did not rescue them, giving -0.156, -0.075 and -0.053 respectively. The null therefore survives its strongest objection.

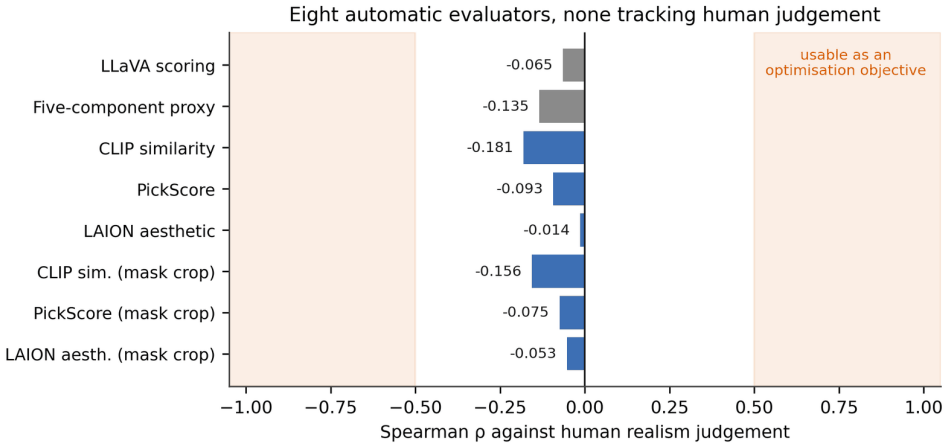


Figure 4.1: Correlation between automatic evaluators and human realism judgement, on paired differences. Shaded regions mark the region an evaluator would need to reach to serve as an optimisation objective.

This result determines a design decision rather than merely documenting a limitation. Because no validated metric was found for this task, `MIPROv2` and `BootstrapFewShot` were not considered methodologically justified: both require a metric for scoring candidate outputs. By contrast, `LabeledFewShot` requires only labelled examples, so it can be used without claiming that an unvalidated automatic score represents human perception. The choice of optimiser in Section 3.2.6 follows from the measurement reported here.

4.2 Human Evaluation as the Primary Evidence

The failure of automatic evaluators did not mean that the editing system could not be evaluated; it meant that the evaluation target had to return to the people whose judgement

the automatic models were intended to approximate. Human ratings were therefore treated as the primary evidence for realism, identity preservation and instruction following.

This also changed the role of the dataset. Rather than using human scores only as a final validation set, the collected annotations became evidence for understanding where the system succeeds and fails. The two prompt-rewriting studies below were therefore designed as blind paired comparisons in which the source image and generation settings were controlled and human judgement determined whether rewriting produced a meaningful difference.

4.3 Study 1: Rewriting Against Hand-Authored Prompts

The first study asked whether a language model could improve on the hand-authored positive prompts of the earlier edit registry, under the protocol of Section 3.3.1. 90 pairs were rated for realism by six annotators (540 realism ratings). Rewritten prompts were rated 0.57 points higher on the five-point realism scale (3.02 against 3.59; 95% CI $[0.26, 0.88]$; Wilcoxon $p = 0.001$, paired standardized effect size $d = 0.43$ [29]). At the pair level, the rewrite had higher mean realism in 62.2% of the 90 pairs. In the separate forced-choice component, it was preferred in 169 of 270 judgements (62.6%, $p < 0.001$). Figures 4.2 and 4.3 show the eight largest improvements, and then four of them at a size where the difference can actually be inspected.

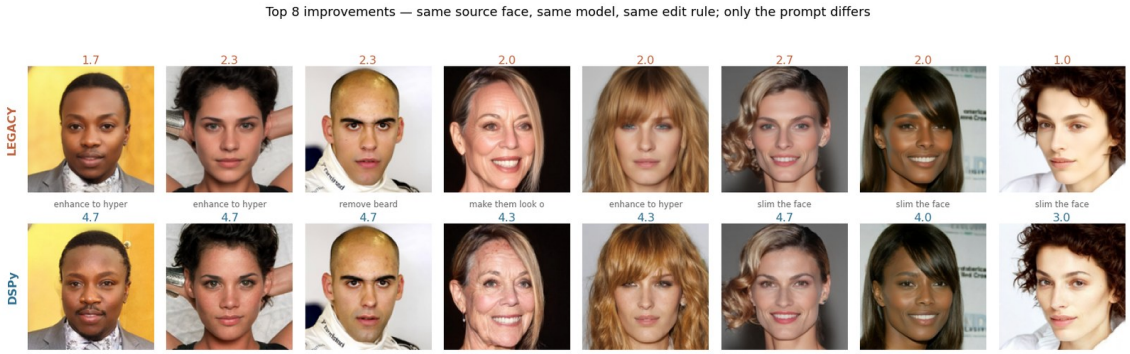


Figure 4.2: Study 1, the eight largest improvements. Top row, the registry prompt; bottom row, the rewrite. Same source face, model, mask and edit rule in every column, so only the positive prompt differs. Mean human realism is printed above each panel.

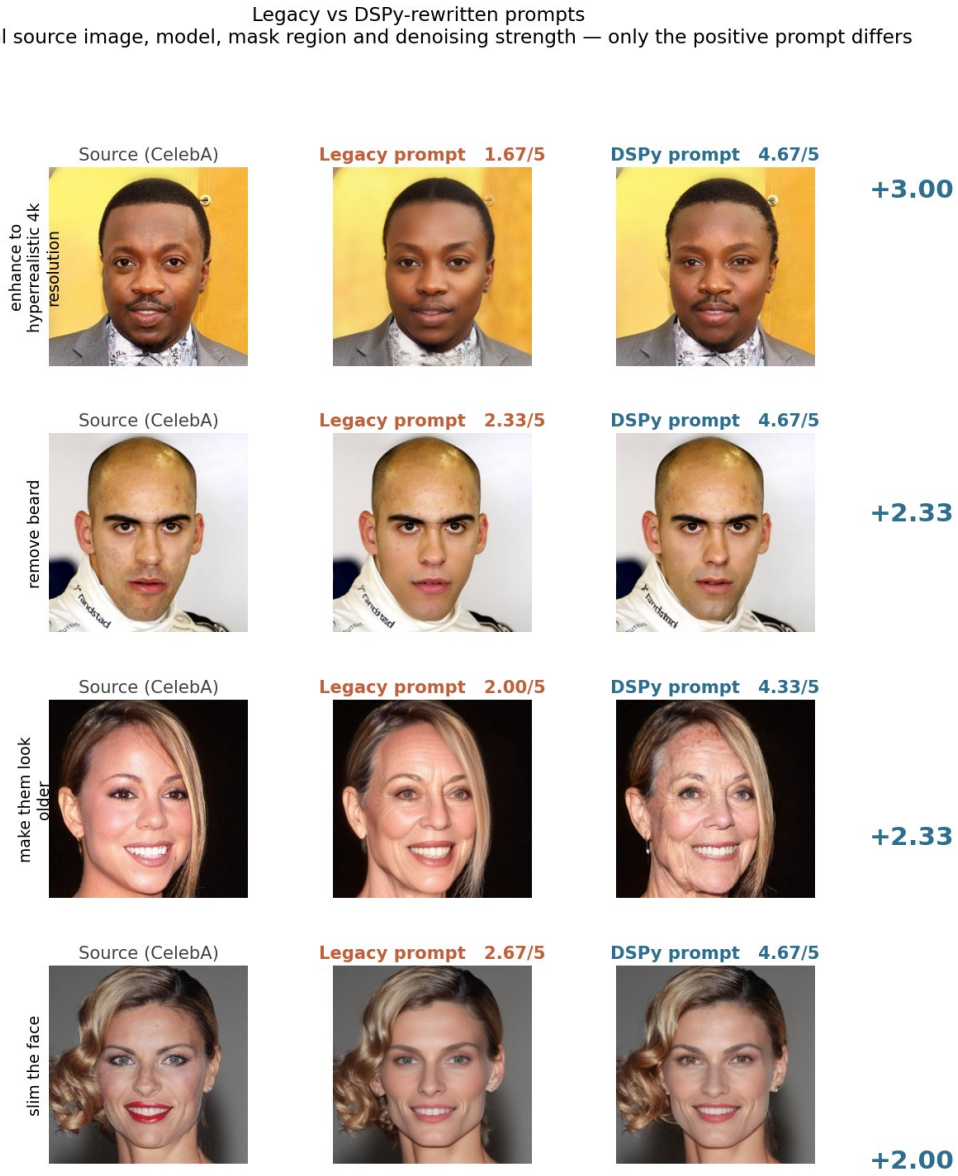


Figure 4.3: Study 1, four of those cases at full size, with the source photograph for reference. Identical source image, diffusion backend, mask region, seed and denoising strength; only the positive prompt differs.

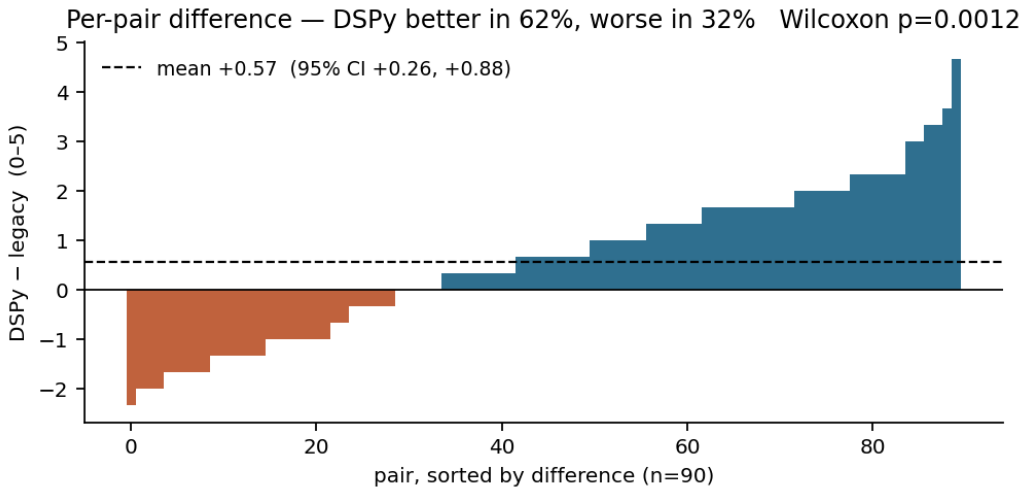


Figure 4.4: Study 1, per-pair difference in mean human realism, sorted. Rewritten prompts were rated higher in 62.2% of pairs and lower in 32%.

All six annotators rated the rewritten arm higher on average, with per-annotator differences from +0.29 to +0.89 (Figure 4.5), so the effect is not carried by a single rater. The same figure is a reminder of why scores are centred within annotator before aggregation, since the annotator means themselves span more than a point.

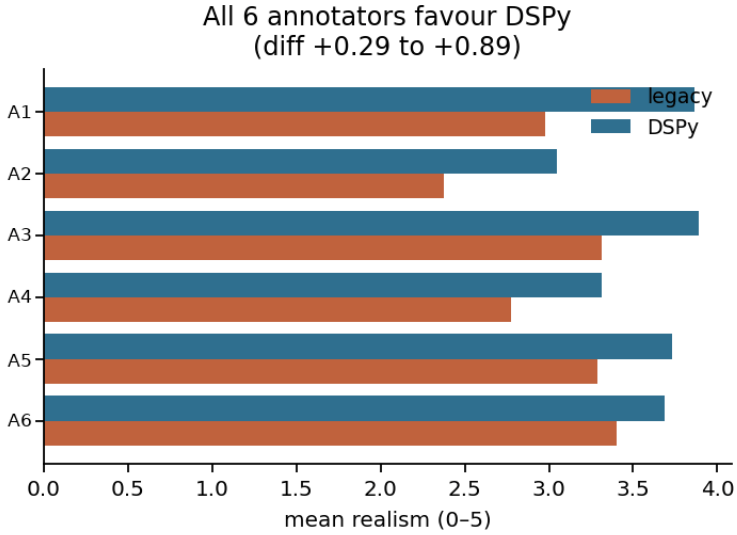


Figure 4.5: Study 1, mean realism per annotator under both prompt sets. Annotators are anonymised. Every one of the six rated the rewritten arm higher, and the spread between annotator means is itself larger than the effect being measured.

4.3.1 The Gain Is Concentrated in the Weakest Prompts

The gain was not uniform. It was concentrated among prompts that scored worst under the original wording. The only per-category contrast to survive Holm correction was `slim the face`, which rose from 1.63 to 3.30 (+1.67, Holm $p = 0.018$); `remove beard` rose

similarly (2.00 to 3.53); while already-strong prompts such as `add beard` (3.80) barely moved, and `add full glamour makeup` moved slightly the wrong way. Figure 4.6 plots improvement against the original score, and the negative slope (-0.72, $r = -0.69$) is the shape of that pattern. Figure 4.7 gives the same result category by category.

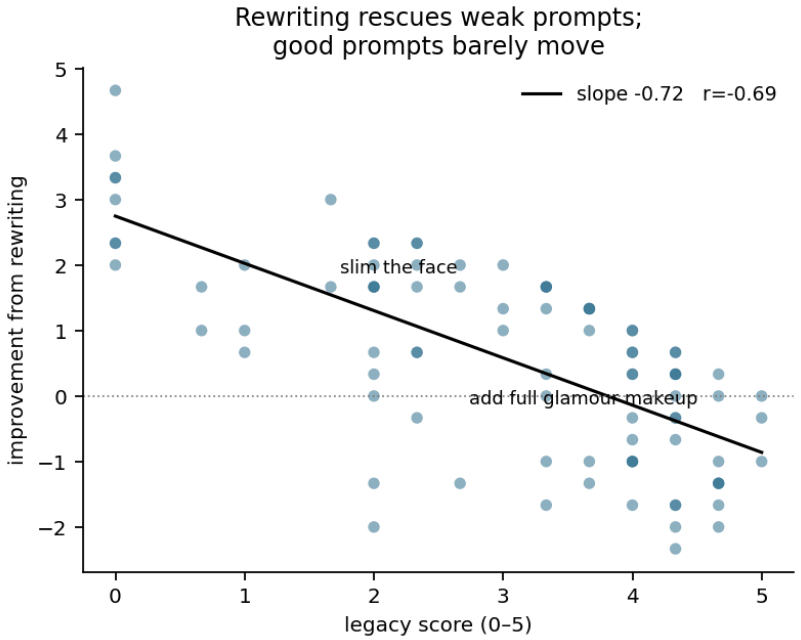


Figure 4.6: Study 1, improvement from rewriting against the original prompt's score. The negative slope indicates that rewriting repairs weak prompts and leaves strong ones roughly unchanged.

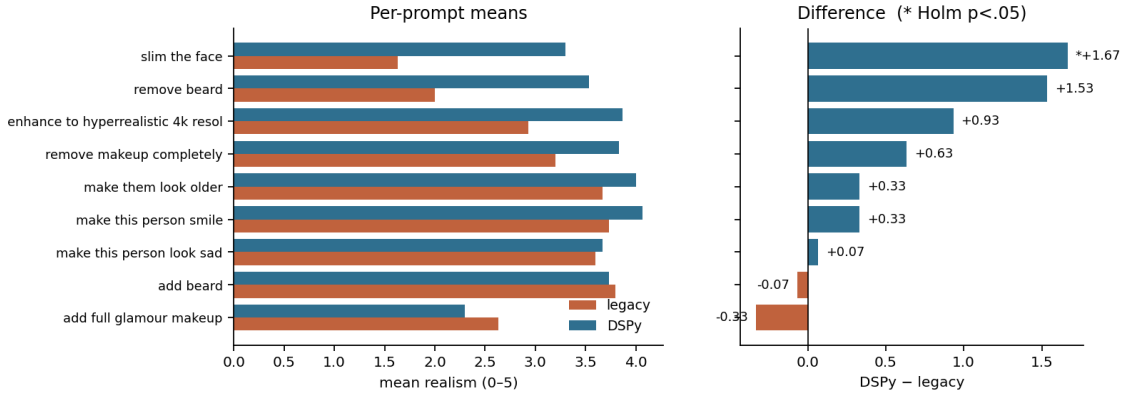


Figure 4.7: Study 1, mean human realism by edit category under both prompt sets. The asterisk denotes Holm-corrected $p < 0.05$ (`slim the face`).

That shape is the first hint of the account this report ends with. A rewriter that lifts a 1.63 prompt to 3.30 and leaves a 3.80 prompt alone is not adding quality. It is removing a defect, which means the size of the measured gain depends almost entirely on how defective the baseline was.

4.4 Why a Second Study Was Necessary

Table 4.3 lists the prompt pairs from Study 1, and reading them raises an obvious objection. The registry prompts are long and internally repetitive. The `slim the face` baseline states `same gender` twice, `natural skin texture` twice and `visible pores` twice within one prompt, and separately asserts `preserve identity` and `preserve facial features`. Seven of the nine baselines end with the same eight-tag boilerplate regardless of what the edit is. A rewriter that improves on that has demonstrated that it can tidy a poorly written prompt. It has not demonstrated that it can improve on what a person would actually type.

The comparison column makes the mechanism visible. The rewrites are shorter, drop the duplicated tags and the hyperbolic vocabulary, and in the two removal cases they describe the end state rather than naming the thing being removed. That last property is incidental in Study 1 and turns out to be the entire effect in Study 2. The table is printed at the end of this chapter, since it needs a full page.

Study 2 replaces the baseline accordingly. Arm A becomes the user's own free-text request plus the shared identity anchor, under the protocol of Section 3.3.2.

4.5 Study 2: Rewriting Against the User's Own Wording

Across 240 annotation tasks covering 172 unique pairs, rewriting produced no overall improvement on any dimension (Figure 4.8, Table 4.1). The rewrite was preferred in 80 of 168 decided comparisons, 47.6% (95% CI 39.9%–55.5%, binomial $p = 0.589$), with 72 ties.

Table 4.1: Study 2 primary outcomes. Arm A is the user's own wording, arm B its DSPy rewrite. Scores are on a 1–5 scale and $\Delta = B - A$. p values are from annotation-level Wilcoxon signed-rank [30] tests, Holm-corrected across the three outcomes. The 240 annotation tasks cover 172 unique pairs, with 68 pairs rated twice.

Outcome	n	A	B	Δ (95% CI)	d_z	p_{Holm}
Realism	240	3.65	3.57	-0.08 $[-0.21, +0.05]$	-0.08	0.337
Identity preservation	240	3.53	3.41	-0.12 $[-0.27, +0.03]$	-0.10	0.337
Edit compliance	240	3.23	3.40	$+0.17$ $[-0.02, +0.36]$	$+0.11$	0.337

The contrast with Study 1 is itself a result. Once the weak hand-authored baseline was replaced with ordinary user wording, the apparent general advantage of rewriting disappeared. This shows that the benefit observed in Study 1 cannot be interpreted as a universal property of DSPy rewriting.

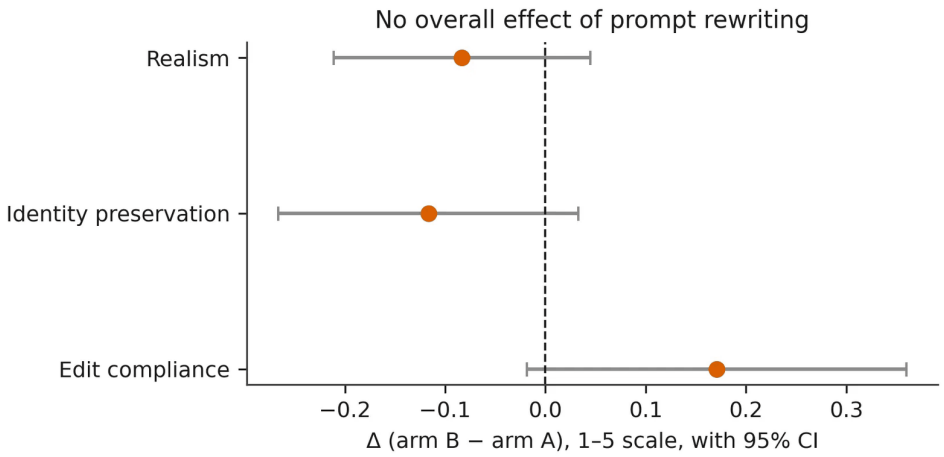


Figure 4.8: Study 2, mean paired difference on each rated dimension with 95% confidence intervals. All three intervals contain zero.

Three properties of the measurement matter for interpreting this null. Because 68 of the 172 unique pairs received a second annotation, the 240 annotation tasks are not all independent image pairs. We therefore do not use the earlier minimum-detectable-difference calculation as evidence of equivalence; the confidence intervals are the appropriate summary of the observed uncertainty. Inter-annotator reliability was ICC(1) = 0.41 [31] for realism, 0.51 for identity and 0.54 for compliance. On the conventional bands these sit in the poor-to-moderate range rather than the good one, which we state plainly rather than describing them as adequate. They are what subjective aesthetic judgement of near-identical image pairs produces, and the two raters of a doubly-judged pair agreed on the forced-choice preference exactly half the time against a chance rate of one third across the three response options (rewrite, original, or tie). Per-annotator mean ratings spanned 2.07 points, confirming the leniency effect that motivated centring scores within annotator.

One exploratory check is worth recording because it points the other way. Restricting to the 68 pairs carrying two raters, where averaging reduces measurement noise, the rewrite is worse on two dimensions, realism -0.18 ($p = 0.042$) and identity -0.25 ($p = 0.013$), with compliance unchanged at $+0.12$ ($p = 0.456$). This subgroup was not pre-specified and these p values are uncorrected, so it does not overturn the primary result. It does mean the null should be read as a null or a small negative rather than as evidence that the two arms are equivalent.

4.5.1 Effect by Input Register

The four registers behave differently (Figure 4.9). Only `bare` single-word requests are significantly affected, and the effect is negative: realism -0.37 with the rewrite preferred in just 25.0% of decided comparisons (Holm-corrected $p = 0.0038$, the only register to survive correction). The `bare` condition also has the *highest* arm-A realism of any register at 3.95, which is informative: a minimal prompt produces a small conservative edit, and elaborating it into eight or ten tags pushes a stronger change than the request warranted.

The Kruskal–Wallis test across registers gives $H = 6.96$, $p = 0.073$, so the register pattern is suggestive rather than established.

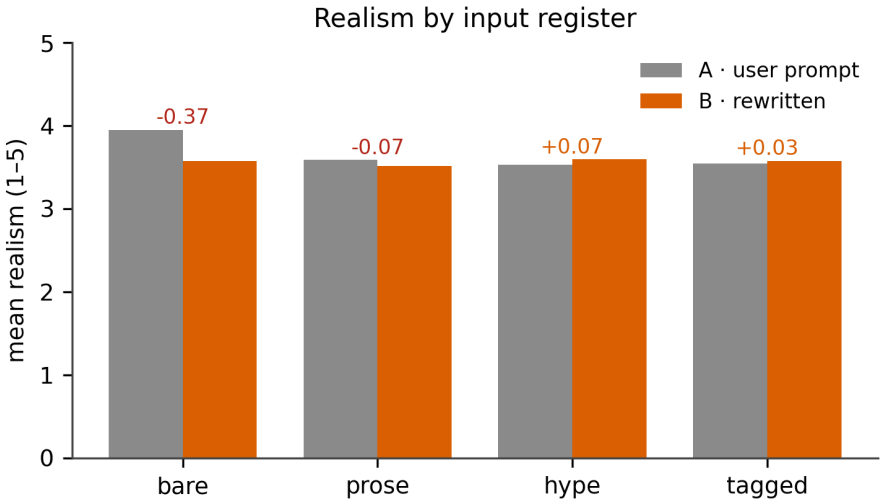


Figure 4.9: Study 2, mean realism by input register under both arms, with the paired difference printed above each pair. Only single-word requests are significantly affected, and negatively.

4.6 Where Rewriting Does Help

The aggregate null becomes more informative when the requests are separated by what the rewriting has to do. For requests that ask to remove a discrete visual element, makeup or facial hair, rewriting improves every dimension substantially; for everything else it does not (Table 4.2, Figure 4.10).

Table 4.2: Study 2, requests that remove a discrete visual element against all others. p values are Mann–Whitney between the two groups [32].

Group	n	Δrealism	Δidentity	Δcompliance	B preferred
Remove a discrete element	29	+0.45	+0.17	+1.31	70.8%
All other requests	211	−0.16	−0.16	+0.01	43.8%
Between-group p		0.0044	0.077	0.0005	

Within the removal group the rewrite was preferred in 17 of 24 decided comparisons, 70.8% (95% CI 48.9%–87.4%), and the compliance gain is significant within group by Wilcoxon ($p = 0.0022$). Both constituent categories move independently in the same direction: `beard_remove` gains +1.43 on compliance ($p = 0.027$, preferred 83.3%) and `remove_makeup` gains +1.38 ($p = 0.006$, preferred 66.7%), so the effect is not carried by one edit type.

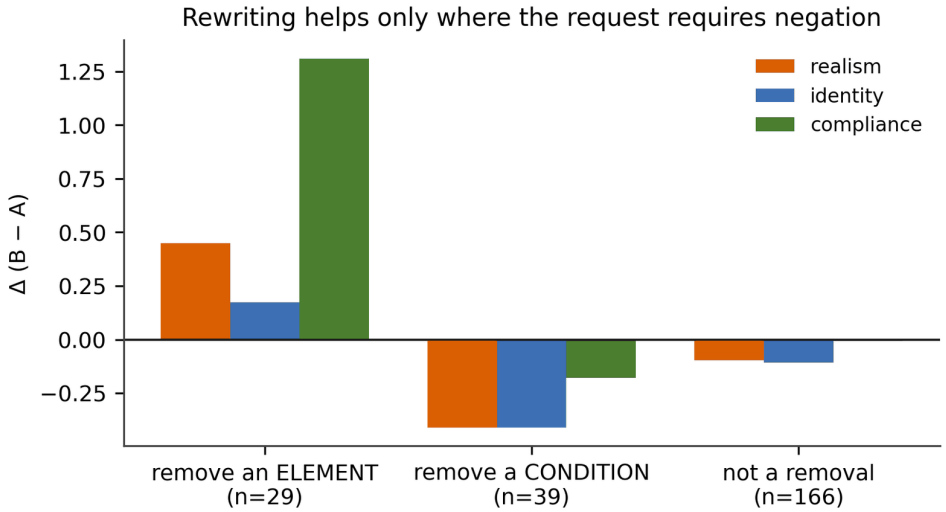


Figure 4.10: Study 2, paired differences for 29 discrete-element removals, 39 skin-condition removals, and 166 non-removal requests. Six additional removal requests are included in the "all other" group of Table 4.2 but are not displayed in this three-way figure because they do not belong to either routed subgroup.

4.6.1 The Mechanism

The explanation follows from how the diffusion backend consumes text. The observed pattern is consistent with a known difficulty in text-conditioned image generation: negative wording such as "remove beard" still contains the visually salient concept "beard", and the model may respond to that concept rather than reliably representing its absence. In our experiment, describing the desired end state instead produced substantially better results for removal of discrete visual elements. A rewrite that describes the resulting state, `clean shaven jaw, bare skin, faint shadow, natural pores`, expresses the same intention without naming the attribute, and the conflict disappears. Figures 4.11 and 4.12 show the two largest instances.

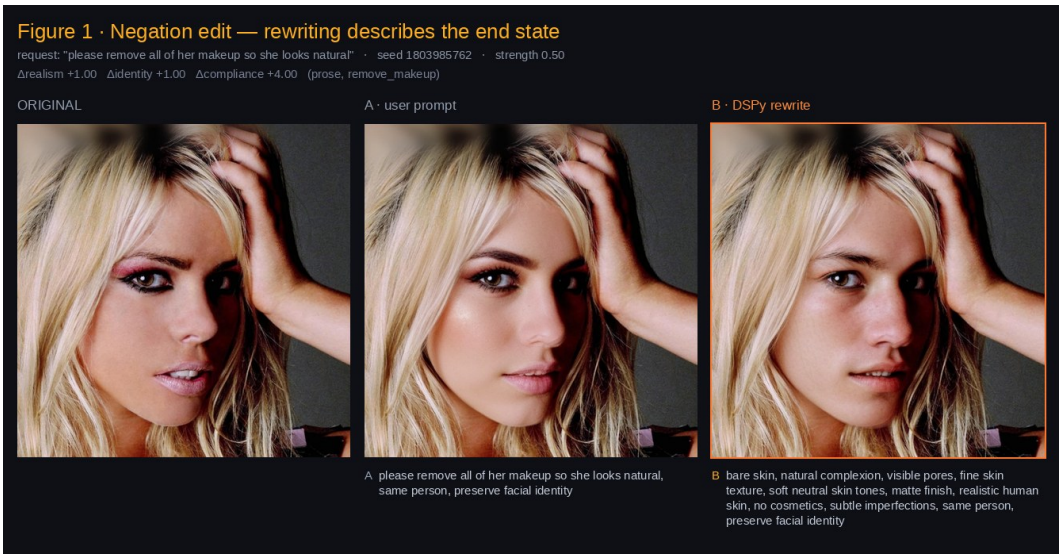


Figure 4.11: A removal request. Arm A repeats the user's wording, which names the makeup being removed; arm B describes the bare skin that should remain. Identical source, mask, seed and strength.

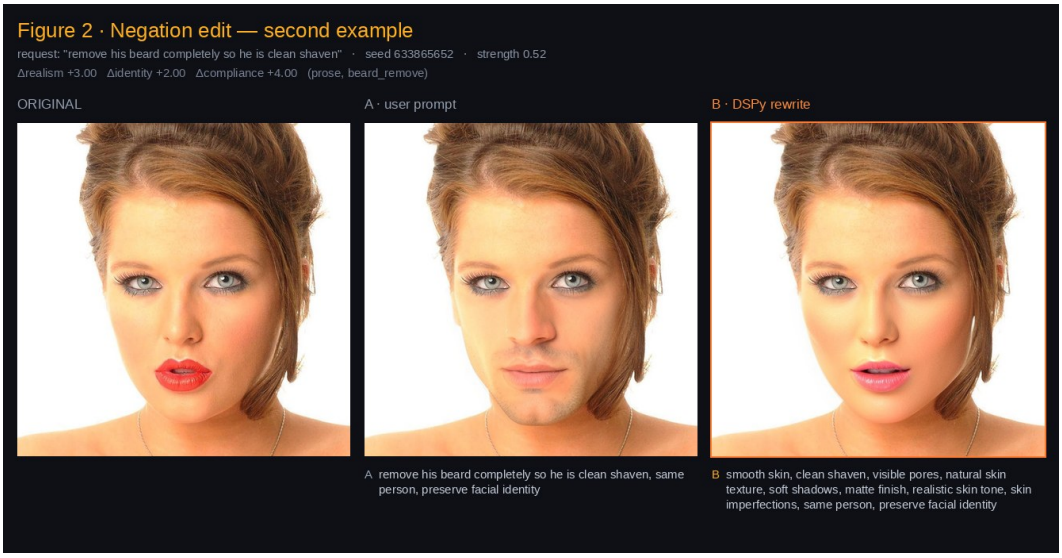


Figure 4.12: A second removal request, showing the same pattern on facial hair.

4.6.2 A Differential Prediction

The mechanism makes a prediction that could have failed. If the benefit comes from avoiding an injected object concept, then removing a skin *condition*, acne, dark circles, fine lines, should not benefit, because there is no strong object concept being wrongly introduced in the first place. Those requests are phrased as removals in exactly the same way.

They do not benefit. Requests removing a skin condition score Δrealism -0.41, Δidentity -0.41, Δcompliance -0.18, with the rewrite preferred in only 16.0% of decided comparisons, against $+0.45$, $+0.17$, $+1.31$ and 70.8% for removal of a discrete element.

The contrast is significant on realism and compliance ($p = 0.0017$ for both) and on identity ($p = 0.042$). Figure 4.13 shows a representative case, in which the rewrite elaborates a one-word request into a full skin-quality description and the result is rated worse on all three dimensions.

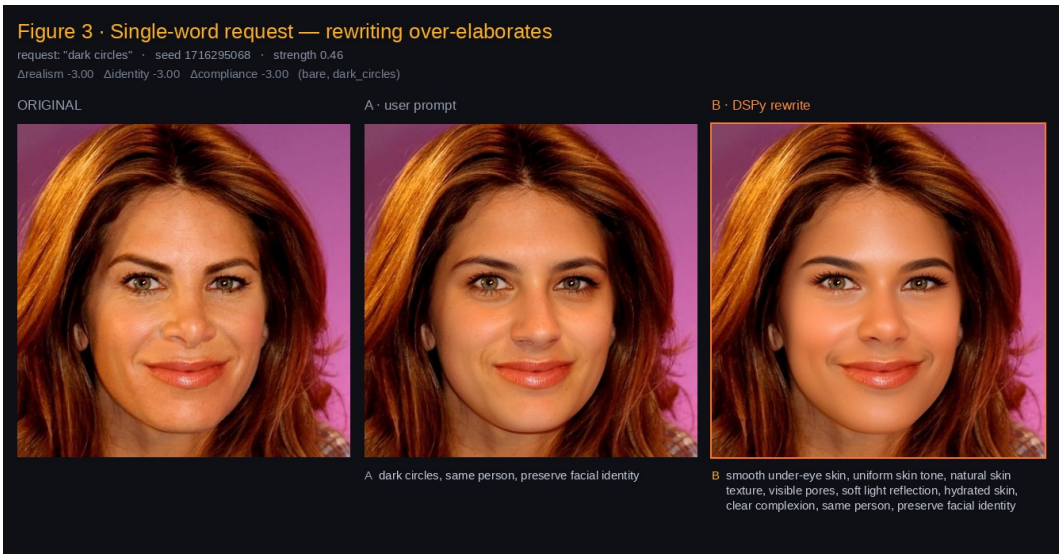


Figure 4.13: A skin-condition request phrased as a removal. The rewrite elaborates the request rather than translating it, and all three ratings fall.

As an exploratory check against simple post-hoc category selection, we enumerated all $\binom{15}{2} = 105$ ways of selecting two of the fifteen edit categories and ranked them by total gain. The pair selected by the mechanism ranks first (permutation $p = 0.0095$); the next-best pairs, {`older`, `remove_makeup`} and {`nose`, `remove_makeup`}, each contain one of the same two categories.

4.6.3 Verbosity

Rewriting roughly doubled prompt length, from a mean of 12.6 words to 26.8. Across all judgements, the number of words added correlates negatively with the change in realism ($\rho = -0.165$, $p = 0.011$) and in identity ($\rho = -0.183$, $p = 0.004$), and the pattern is monotone across quartiles of words added (Figure 4.14): the lightest quartile gains +0.24 realism and +0.14 identity, the heaviest loses −0.37 and −0.46.

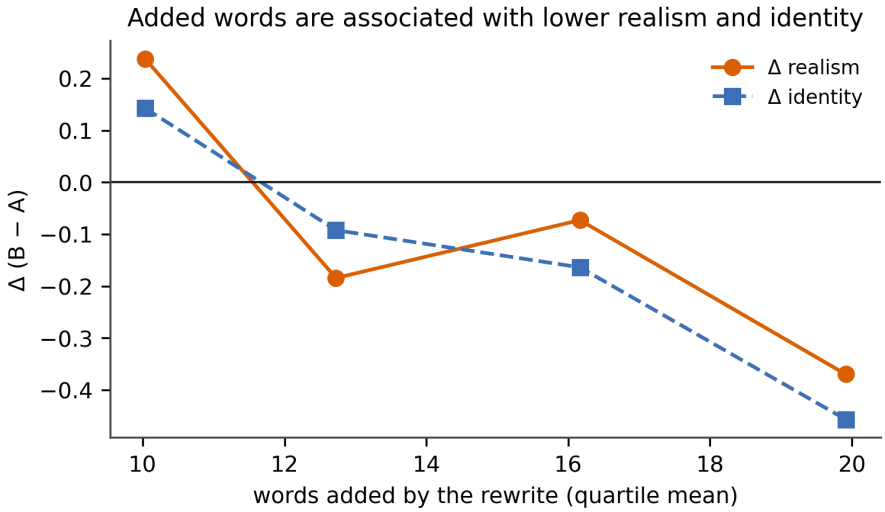


Figure 4.14: Study 2, paired differences by quartile of words added by the rewrite.

This association should be interpreted with an important caveat. Restricted to non-removal requests, the correlations attenuate to $\rho = -0.127$ ($p = 0.066$) for realism and $\rho = -0.116$ ($p = 0.094$) for identity, and no longer reach conventional significance. Removal requests happen to receive shorter rewrites, so length is partly a marker of which requests were being translated rather than elaborated. The verbosity association is therefore suggestive; the negation effect is not.

Finally, the two effects are not a trade-off. Change in compliance correlates *positively* with change in realism ($\rho = +0.286$, $p < 10^{-5}$) and with change in identity ($\rho = +0.245$, $p = 0.0001$), so a rewrite either improves both or degrades both. Compliance was not purchased at the cost of realism.

4.6.4 Per-Category Behaviour

Figure 4.15 orders the fifteen edit categories by change in realism. Interpretation should be cautious: n per category is 10–24 and no correction is applied, so some categories will appear extreme by chance. The ordering is nonetheless consistent with the account above, with the two removal-of-element categories at the positive end and `slimming` and `dark circles` at the negative end.

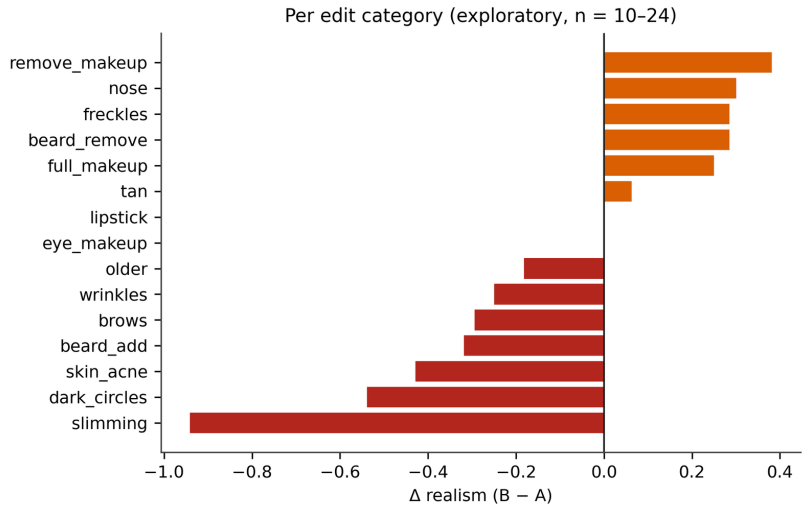


Figure 4.15: Study 2, paired difference in realism by edit category. Exploratory; $n = 10$–24 per category, uncorrected.

4.7 Identity Preservation

Each of the 240 judgements rated both arms, giving 480 image-level identity ratings, and these allow the identity question to be answered by measurement rather than assertion. Identity ratings were more strongly associated with how much of the face is regenerated ($\rho = -0.427$ between mask coverage and identity rating, $p < 10^{-21}$) and much more weakly to how strongly it is regenerated ($\rho = -0.148$ with denoising strength, $p = 0.001$). Within a coverage band the strength relationship disappears entirely: among masks covering less than 12% of the frame the correlation is +0.029 ($p = 0.75$), and among larger masks −0.073 ($p = 0.17$). Small masks score 4.36 on identity against 3.16 for large masks. Figure 4.16 shows both relationships.

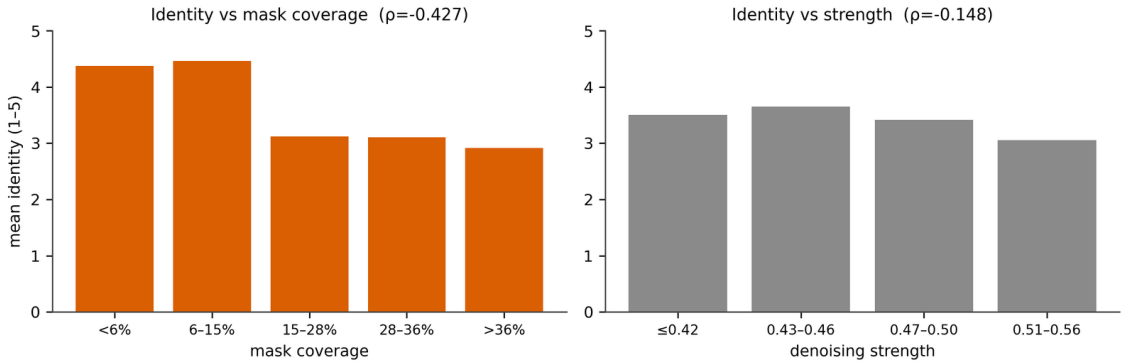


Figure 4.16: Mean identity rating by mask coverage and by denoising strength, both arms pooled. Identity ratings showed a stronger association with mask coverage than with denoising strength.

The practical consequence in this dataset is that restricting the mask is more strongly associated with identity preservation than lowering the denoising strength. Broad edits

should therefore be treated as higher identity-risk cases, while the present data do not establish a universal causal threshold. The system therefore reports an identity-risk estimate from the measured coverage rather than silently resolving the trade-off.

4.7.1 A Correction We Had To Make

An important result of the analysis was not only what the coverage correlation revealed, but also the correction of an initial interpretation based on that correlation. This distinction is important because it separates an observed association from an unsupported system-design rule. Coverage also correlates with compliance at $\rho = -0.026$ ($p = 0.58$), that is, not at all. We initially read the pair of correlations together as implying that masks could be shrunk to protect identity at no cost to the edit, and reduced the face-wide masks accordingly. The edits degraded noticeably: makeup removal, beard removal and de-ageing all stopped working, and the change was reverted.

The error is instructive. That near-zero correlation is measured *across* edit categories, and category is a confounder. Small-mask categories such as lipstick and brows are easy edits that score well on compliance; large-mask categories such as makeup removal and tanning are hard edits that score poorly. Pooled together the two effects cancel, producing a correlation near zero that says nothing about the within-category question of what happens when a given edit's mask is shrunk. There the answer is direct: removing makeup from two thirds of a face leaves a third of the makeup. The identity finding stands; the inference drawn from it did not.

4.8 Discussion

Taken together the two studies support a narrower and more useful claim than either does alone. Prompt rewriting is not a general improvement to a diffusion editing pipeline. It is a repair for prompts that are poorly formed, which is what Study 1 measured, and a translation that can help a specific class of removal request, which is what Study 2 isolated. Where the request is already a reasonable description of a target image, rewriting adds words without adding information, and the measured effect is null or slightly negative.

The two baselines explain the two headline numbers. Study 1's +0.57 is real, but its size is a property of the registry prompts rather than of the rewriter, as Figure 4.6 shows directly. The worse the baseline, the larger the gain, and a baseline at 1.63 out of 5 leaves a great deal of room. Replace that baseline with a plain-language request and the average gain disappears, which is the honest answer to the question a deployed system actually faces.

What survives the change of baseline is the negation effect, and it survives because it is not a tidying operation at all. Asking the model to remove something by name is a request it handles unreliably, and describing the end state instead provides a representation that avoids naming the unwanted visual concept. That is why the effect concentrates so cleanly

in that group and nowhere else, and why the differential prediction on skin conditions held.

That distinction has a direct design consequence. Because rewriting helps decisively in one identifiable case and is neutral to harmful in others, and because the evidence for the conditional effect rests on 29 judgements, the deployed system does not gate on it. As implemented in the rewriting workflow, it rewrites every request, presents both versions and records which the user keeps. That preserves the option the study cannot yet confidently resolve, and each choice is itself a blind paired comparison with randomised presentation order, so ordinary use continues to accumulate evidence.

The main limitations are the size of the removal subgroup (29 annotation tasks, adequate for the observed compliance effect but limited for realism, where the within-group $p = 0.082$), the exploratory status of the per-category, register and doubly-rated analyses, the fact that only 68 of 172 pairs carried two raters, and the participation of both authors among the six Study 1 annotators. The negation finding warrants a prospective confirmation, and the effect sizes reported here are the predictions such a study would test.

4.8.1 A Qualitative Extension: Transformation Into Infancy

Section 3.4 of the methodology describes an editing mode that departs from every other request the system handles: instead of holding the subject's bone structure fixed, it asks the model to replace it with infant proportions, on the reasoning that a face's identity-bearing geometry is exactly what a baby version of that face cannot share. No annotated dataset exists for this mode, so nothing here is reported as a statistic; it is a qualitative observation about where the approach currently falls short.

The transformation reads convincingly for source faces that are already youthful. It reads less convincingly the further the source is from that range, an older teenager or an adult, where the gap between the input geometry and plausible infant geometry is largest and the model has the most work to do inside a single inpainting pass. The result in those cases can look like an approximation of a baby rather than a specific baby version of the person in the photograph, particularly around the lower face. We do not have a way to state precisely when this happens, because doing so would need exactly the kind of measurement that Section 4.1 shows is not currently available: there is no automatic way to score whether a given output is a plausible infant version of a given adult, any more than there was an automatic way to score whether a rewritten prompt was a good one. The honest path forward is the one this report has already taken once, a blind protocol in which people who have not seen the source judge whether the result is a credible match, rather than an automated proxy for that judgement.

Table 4.3: Study 1, hand-authored Vision Studio prompts against their DSPy rewrites, with mean human realism (1–5) from the blind paired evaluation ($n = 10$ pairs per edit, 6 annotators each). * Holm-corrected $p < 0.05$. The repetition visible in the left-hand column is what motivated the harder baseline of Study 2.

Edit	Legacy	DSPy	Δ	Legacy prompt	DSPy rewrite
slim the face	1.63	3.30	+1.67*	slim narrow face, sharp defined jawline, high cheekbones, natural skin texture, same gender, same person, exact same gender, preserve identity, preserve facial features, natural skin texture, visible pores, realistic portrait photo, photorealistic	slimmer face, sharper jawline, higher cheekbones, natural skin texture, visible pores, soft skin, same person, same gender, preserve facial features
remove beard	2.00	3.53	+1.53	clean shaven smooth skin, no facial hair, natural skin texture, visible pores, same gender, same face structure, same person, exact same gender, preserve identity, preserve facial features, natural skin texture, visible pores, realistic portrait photo, photorealistic	clean shaven skin, smooth texture, visible pores, natural skin, no beard, face structure preserved, realistic portrait, same gender, same person, preserve facial features
enhance to hyperrealistic 4k resolution	2.93	3.87	+0.93	ultra realistic hyperdetailed portrait, 8k RAW photo, sharp natural skin texture, individual hair strands visible, pores, same gender, same person, exact same gender, preserve identity, preserve facial features, natural skin texture, visible pores, realistic portrait photo, photorealistic	ultra detailed skin texture, fine hair strands, visible pores, realistic lighting, same person, same gender, preserve facial features
remove makeup completely	3.20	3.83	+0.63	completely bare natural skin, no makeup at all, fresh faced, visible pores, same gender, same person, exact same gender, preserve identity, preserve facial features, natural skin texture, visible pores, realistic portrait photo, photorealistic	bare natural skin, no makeup, visible pores, realistic portrait, same person, same gender, preserve facial features
make them look older	3.67	4.00	+0.33	older aged appearance, natural wrinkles, age spots, realistic aging skin texture, same gender, same facial structure, same person, exact same gender, preserve identity, preserve facial features, natural skin texture, visible pores, realistic portrait photo, photorealistic	natural wrinkles, age spots, weathered skin, sagging cheeks, jowls, realistic aging, visible pores, same person, same gender, preserve facial features
make this person smile	3.73	4.07	+0.33	natural warm smile showing teeth, happy expression, natural skin texture, visible pores, same gender, same face structure, same person, exact same gender, preserve identity, preserve facial features, natural skin texture, visible pores, realistic portrait photo, photorealistic	soft warm smile, showing teeth, happy expression, natural skin texture, visible pores, subtle wrinkles, same gender, same face structure, preserve identity, preserve facial features
make this person look sad	3.60	3.67	+0.07	sad expression, frowning, upset, natural skin texture, visible pores, same gender, same face structure, same person, exact same gender, preserve identity, preserve facial features, natural skin texture, visible pores, realistic portrait photo, photorealistic	deep furrows, downturned lips, drooping eyes, sorrowful gaze, melancholic expression, visible pores, soft skin texture, subtle shadows, natural lighting, same person, same gender, preserve facial features
add beard	3.80	3.73	−0.07	thick realistic beard, dense facial hair, natural skin texture, visible pores, same gender, same face structure, same person, exact same gender, preserve identity, preserve facial features, natural skin texture, visible pores, realistic portrait photo, photorealistic	thick beard, dense facial hair, natural skin texture, visible pores, rough texture, stubble, facial hair growth, same gender, same face structure, preserve identity, preserve facial features
add full glamour makeup	2.63	2.30	−0.33	full face professional makeup, heavy foundation, contouring, bold lipstick, smoky eye, same gender, same face structure, same person, exact same gender, preserve identity, preserve facial features, natural skin texture, visible pores, realistic portrait photo, photorealistic	full glam makeup, bold foundation, dramatic contour, bright bold lipstick, striking smoky eye, same gender, same face structure, preserve identity, preserve facial features

Chapter 5

Impacts of the Project

This chapter sets out what Vision Studio does and does not do for the people it affects, covering societal and health consequences, ethical constraints on its use, and its computational cost.

5.1 Societal and Health Impact

A cosmetic preview changes what someone believes is possible for their own face, so two of the findings in Chapter 4 matter beyond the pipeline. The first is that identity ratings are more strongly associated with how much of the face is regenerated than with how strongly it is regenerated. In the evaluated setting, broader masks therefore correspond to greater identity risk. A user looking at such a preview is partly looking at someone else. The system therefore reports an identity-risk estimate derived from measured mask coverage instead of resolving that trade-off quietly on the user's behalf. The second is that identity preservation here is measured against 480 human ratings rather than asserted through a similarity score; this provides direct perceptual evidence for the evaluated sample, rather than a claim that a single similarity metric is sufficient.

Demographic bias in generative imagery is documented. Wiegand et al. [3] found substantial demographic inaccuracies and an over-representation of White and normal-weight individuals in 9,060 AI-generated patient images. That study covers generation rather than editing, so it does not directly establish bias in Vision Studio. It nevertheless motivates attention to demographic representation in generative-image systems. Reducing that gap was among the motivations for this project. We did not reduce it. The pipeline records no demographic attributes, no fairness analysis was run, and the CelebA source pool [27] carries its own skew, so any audit performed on the current dataset would inherit that skew rather than correct for it. The objective remains stated rather than demonstrated, and it is listed as unfinished work in Section 8.2.

What the project does contribute towards a later audit is instrumentation. Every generated pair records the source image, both prompts, the full mask specification, the random seed and the model settings, and an image can be regenerated under the same

recorded model and generation configuration. A fairness analysis can therefore be run over the collected data, or over a re-sampled source set, without repeating the generation work. That property was already used once in practice, to rebuild the image set for the metric-validation experiment after a session reset.

One limit is worth stating plainly because users are unlikely to infer it. Vision Studio is not a medical device and its output is not a prediction of a clinical or surgical result. An inpainted preview of clearer skin says nothing about whether any particular treatment would produce that skin on that person. The system is a visualisation tool, and treating it as a forecast would be a misuse of it.

The authors consider the project broadly aligned with UN Sustainable Development Goal 9 through a reproducible method that runs on commodity hardware, and related to SDG 10 in intent only, subject to the audit described above.

5.2 Ethical Considerations

Cosmetic-editing technology can narrow what people accept as an ordinary face. Vision Studio limits that in three ways. Edits are confined to a landmark-composed mask, so pixels outside the mask are preserved; hair, clothing and background are unchanged when they fall outside the resolved mask. No generated version is silently selected on the user's behalf: the system renders both the user's own wording and the rewrite and leaves the selection to the person. And the rewriting stage, which Chapter 4 shows is neutral or slightly harmful outside one class of request, is not allowed to override the user's phrasing silently.

The system is built for consensual, self-directed use. It is not designed to fabricate images of other people, and the masking scheme does not prevent that use, so the restriction is a stated condition rather than an enforced one. Anyone deploying this pipeline publicly would need consent controls that the research prototype does not have.

Source faces come from CelebA [27], a public research dataset containing photographs of real people. We do not assume that inclusion in a research dataset constitutes consent for arbitrary cosmetic modification by downstream users. The images were used to evaluate the system rather than distributed as a product, and a deployed version should operate on images supplied with appropriate consent and permissions.

Eighteen people scored image pairs across the two studies, six in Study 1 and twelve in Study 2. They judged images only. No personal data was collected from them, participation was voluntary and uncompensated, and the ratings are stored against pair identifiers with an annotator index used solely for within-annotator centring. The per-annotator figure in Section 4.3 is anonymised for the same reason. Two of the six Study 1 annotators were the authors of this report, which Section 3.3.1 states and Section 4.8 lists among the limitations.

5.3 Environmental Impact

The local diffusion and computer-vision components run on a single Kaggle T4 at roughly 6.5 seconds per generated image; prompt rewriting uses the remote Gemini backend. Study 2 rendered 400 images across 200 pairs, Study 1 a further 180 across 90, with a smaller regeneration run for the metric validation. No backbone was retrained and no reward model was trained, since identity is protected structurally through masking and texture restoration rather than through fine-tuning. Set against preference-alignment work that rests on 137k [4] or 798k [6] human comparisons and the training runs those datasets support, the computational scale of this project is substantially smaller than those large-scale preference-alignment efforts.

Two design choices hold it there. Rewrites are cached, and only successful ones, so repeated requests do not incur repeated language-model calls and a failed call cannot be replayed. Recorded seeds mean that reproducing an earlier result is a regeneration rather than a search.

The metric result in Section 4.1 has an incidental effect in the same direction. Metric-driven optimisers such as `MIPROv2` score unseen outputs, which for this task would mean generating images for every candidate prompt across many trials. Not using that family in the present system also avoids the repeated candidate-generation cost that would be required for metric-driven search.

Chapter 6

Project Planning and Budget

This chapter describes the planning of the project using a Gantt chart and presents the project budget.

6.1 Project Planning

The project is planned across the two capstone terms. The CSE499A phase covers the literature review, the editing engine, the data layer and blind scoring interface, the first evaluation study, and the DSPy rewriting stage with its contract validator. The CSE499B phase covers the rebuild of the masking scheme into composable conditional protection, free-text intent parsing, the generation and annotation of the second study, the metric-validation and identity analyses, the experimental infancy transformation, and the final report. Figure 6.1 shows the schedule.

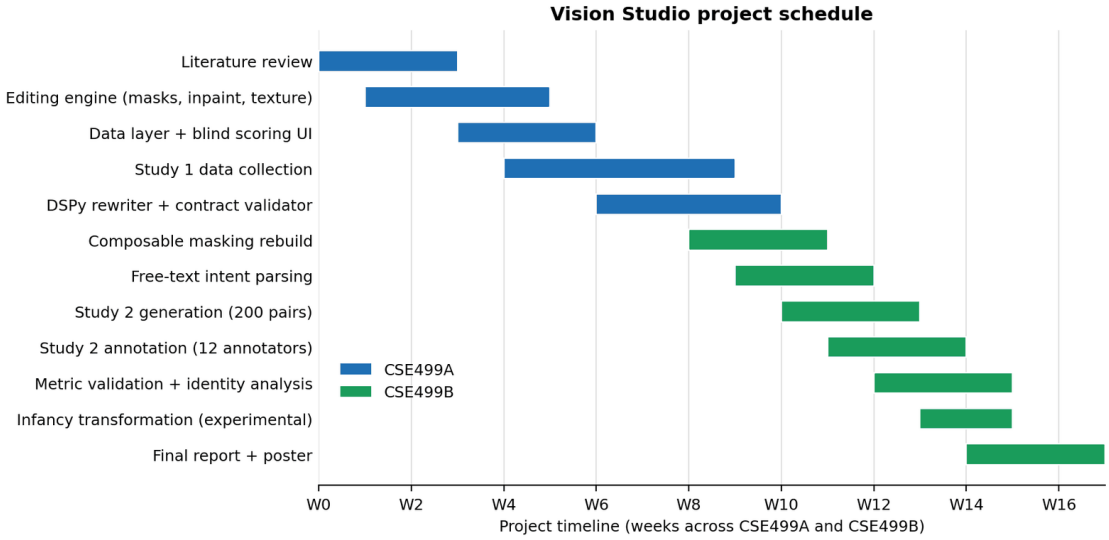


Figure 6.1: Project schedule across the CSE499A and CSE499B terms.

6.2 Budget

Vision Studio is predominantly a software project and relies on free and open-source tools and cloud notebooks with free GPU quotas. The principal costs are optional cloud compute beyond the free tier and a small language-model API budget for the rewriting stage, which ran on a free tier throughout and is carried here only as a contingency. Human annotation was volunteered and is costed at zero, which understates the real cost of the largest input to the project. Table 6.1 summarises the estimate.

Table 6.1: Estimated project budget. All figures in BDT.

Item	**Purpose**	**Estimated cost**
Cloud GPU (Kaggle free quota)	Diffusion generation for both studies	0
Optional extra GPU hours	Faster regeneration when the free quota is exhausted	6000–7000
Language-model API	Prompt rewriting using Gemini	2000–3000
Storage and dataset hosting	CSV archive, generated images, masks	0 (Kaggle)
Miscellaneous	Fonts and design assets for the report and poster	3000–4000
Human annotation	18 annotators, 780 annotation tasks across both studies (540 in Study 1; 240 in Study 2)	0 (volunteered)
Total (approximate)		**15000**

Chapter 7

Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

Table 7.1 maps the complex engineering problem attributes to this project.

Table 7.1: Complex Engineering Problem Attributes for Vision Studio

Attributes	**Description**	**Addressing the CEP in the Project**
P1	Depth of knowledge (K3–K8)	Requires knowledge of computer vision and diffusion models (K3–K4), system and data-pipeline design (K5), engineering/IT tools such as diffusion inpainting, MediaPipe and DSPy (K6), societal and fairness effects of generative imagery (K7), and recent research literature (K8).
P2	Range of conflicting requirements	Edit strength conflicts with identity preservation and realism; a stronger cosmetic change tends to reduce identity retention and skin realism, requiring a balance.
P3	Depth of analysis required	There is no unique solution: mask composition, protection policy, denoising strength, prompt phrasing and evaluation design all admit multiple alternatives, and several were resolved only by controlled experiment rather than by reasoning.
P4	Familiarity of issues	Involves diffusion inpainting, landmark-composed masking, declarative prompt programming, non-parametric paired inference, and the validation of automatic image evaluators against human judgement.
P5	Extent of applicable codes	No standard governs cosmetic-edit quality, so the project defines its own blind human-scoring protocol. Constraints that do apply are the CelebA non-commercial research licence, ordinary data-protection expectations around face imagery, and the IEEE code of ethics on honest reporting of results.
P6	Extent of stakeholder involvement	Stakeholders include end users seeking previews, evaluators who score edits, and the broader community affected by fairness in cosmetic imagery.
P7	Interdependence	The intent parser, mask builder, prompt rewriter, diffusion backend and evaluation harness are interdependent: a change to the mask policy alters what the prompt should describe, and a change to either invalidates the paired comparison unless the remaining variables are held fixed.

7.2 Complex Engineering Activities (CEA)

Table 7.2 maps the complex engineering activity attributes to this project.

Table 7.2: Complex Engineering Activity Attributes for Vision Studio

Attributes	Description	Addressing the CEA in the Project
A1	Range of resources	Uses human resources (evaluators), open datasets, a single GPU, and open-source software, with an optional small API budget.
A2	Level of interactions	Requires coordination among team members, the supervisor, and eighteen volunteer annotators across two evaluation rounds, managed through the blind scoring interface and the CSV data layer.
A3	Innovation	Introduces a composable conditional-protection masking scheme, a declarative DSPy rewriting stage with a deterministic contract validator, two controlled blind paired studies whose contrasting baselines isolate the conditions under which prompt rewriting helps, a mechanism for the conditional effect together with a differential test of it, and an experimental extension of the masking design to a transformation that deliberately departs from identity preservation.
A4	Consequences to society/environment	Measures identity preservation against human ratings rather than asserting it, and reports an identity-risk band to the user. The computational footprint is small, supporting UN SDG 9. The fairness objective that motivated the project is instrumented but not yet audited, so the SDG 10 contribution is stated as intent rather than outcome (Section 5.1).
A5	Familiarity	Requires familiarity with diffusion models, computer-vision tooling, declarative prompt programming, non-parametric paired statistics, and reproducible experimentation.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

Vision Studio is a region-masked diffusion inpainting system for cosmetic face editing in which the wording of the user's request is treated as an engineering problem rather than an input to be passed through unchanged. The editing engine composes a mask from MediaPipe landmarks, protecting a facial part only when the edit does not target it, and restores the original high-frequency skin texture after generation. A DSPy program holds the prompt-writing rules as a declarative signature and rewrites the request before it reaches the generator, with a deterministic validator checking each output against that contract.

We evaluated the rewriting under two blind paired protocols in which the source photograph, mask, random seed, denoising strength and checkpoint were held constant so that the prompt was the only variable. Against the hand-authored prompts of an earlier edit registry, rewriting raised realism by 0.57 points (Wilcoxon $p = 0.001$, $d = 0.43$), with the gain concentrated in the weakest prompts and already-strong ones barely moving. Against the user's own plain-language request, 240 annotation tasks covering 172 unique pairs and twelve annotators, it produced no overall benefit. Realism fell 0.08, identity 0.12, compliance rose 0.17, and none of the three differences survived correction; the rewrite was preferred in 47.6% of decided comparisons. Because 68 pairs received a second annotation, we do not interpret these results through a minimum-detectable-difference claim.

Read together the two studies say something more precise than either does alone. Rewriting repairs prompts that are poorly formed, which is what the first measured, and the size of that gain is a property of the baseline rather than of the rewriter. It does not improve a request that is already a reasonable description of a target image. The exception is systematic and is consistent with a difficulty in handling negated visual concepts. A request to remove something can still activate the concept it names, whereas describing the resulting state avoids naming that concept. In the discrete-element removal subgroup, this translation produced a 1.31-point gain in edit compliance and a 70.8% preference rate. The differential analysis provides a useful check on this interpretation: removing a

skin *condition*, where no discrete object concept is being injected, did not show the same benefit (-0.41 against $+0.45$ on realism, $p = 0.0017$).

We also report two constraints that shape how such systems can be built. Eight automatic evaluators, including mask-cropped variants introduced specifically to test the most plausible objection to the null, failed to track human judgement. The tested metrics were therefore not considered suitable objectives for metric-driven prompt optimisation, and the system is instead compiled from labelled examples. Separately, identity ratings were more strongly associated with how much of the face is regenerated than with denoising strength, which makes mask geometry the more informative design variable in this dataset.

The contribution is therefore a reproducible pipeline whose local diffusion and computer-vision components run on a single commodity GPU, a controlled measurement of what prompt rewriting does and does not do for facial editing, a mechanism that explains the difference and a test of that mechanism, and two negative results that constrain the design of any similar system.

8.2 Future Work

Five directions follow directly from what the studies leave open.

The removal-of-discrete-element finding rests on 29 annotation tasks. It showed a large compliance effect, while the within-group test for realism gives $p = 0.082$. A prospective study restricted to removal requests, with a pre-registered hypothesis and the effect sizes reported here as its predictions, would convert an exploratory result into a confirmed one at modest cost.

The deployed system now presents both versions of every edit and records which the user keeps. Since presentation order is randomised and the source of each option is hidden until after selection, each choice is a paired comparison with randomized presentation order and hidden provenance. Accumulated over normal use this yields preference data that is potentially larger and grounded in an intended-use decision, although such deployment data would still require separate quality controls and consent procedures.

No automatic evaluator tested here measured *identity*. All eight measured realism or general quality. Identity is a full-reference question with a plausible geometric answer, and the landmark guard of Section 3.1.4 already computes one, though it has never been checked against human perception. Validating that residual, or a face-embedding distance, against the 480 identity ratings already collected is the most direct route to the first working automatic evaluator for this pipeline. Such a measure would also close the reward loop, permitting metric-driven DSPy optimisers if a validated evaluator is established.

The infancy transformation described in Section 3.4 has not been evaluated the way everything else in this report has, and it needs to be, for the same reason automatic metrics were ruled out for the main rewriting question: there is no automatic way to score whether an output is a plausible infant version of a specific adult face, so the only honest route is the blind paired protocol already established here, applied to this transformation with its

own annotators and its own baseline conditions.

Finally, the demographic fairness objective that motivated the project remains unaudited. The present system records no demographic attributes, and a fairness analysis across skin tones would require both that instrumentation and a sampling design that guarantees adequate representation within each group. CelebA is not that sampling design, so the audit needs a source set assembled for the purpose rather than the one used here.

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